CHAPTER – V

FREE ELECTRON THEORY OF METALS

5.1 DRUDE-LORENTZ'S CLASSICAL THEORY (FREE ELECTRON GAS MODEL)

Drude, in 1900, postulated that the metals consist of positive ion cores with the valence electrons moving freely among these cores. The electrons are, however, bound to move within the metal due to electrostatic attraction between the positive ion cores and the electrons. The potential field of these ion cores, which is responsible for such an interaction, is assumed to be constant throughout the metal and the mutual repulsion among the electrons is neglected. The behaviour of free electrons moving inside the metals is considered to be similar to that of atoms or molecules in perfect gas. These free electrons are, therefore, also referred to as *free electron gas* and the theory is accordingly named as *free electron gas model.* The free electron gas, however, differs from an ordinary gas in some respects. Firstly, the free electron gas is negatively charged whereas the molecules of an ordinary gas are mostly neutral. Secondly, the concentration of electrons in an electron gas is quite large as compared to the concentration of molecules in an ordinary gas. The valence electrons are also called the conduction electrons and obey the Pauli's exclusion principle. These electrons are responsible for conduction of electricity through metals. Since the conduction electrons move in a uniform electrostatic field of ion cores, their potential energy remains constant and is normally taken as zero, i.e., the existence of ion cores is ignored. Thus the total energy of a conduction electron is equal to its kinetic energy. Also, since the movement of conduction electrons is restricted to within the crystal only, the potential energy of a stationary electron inside a metal is less than the potential energy of an identical electron just outside it. This energy difference, V_o, serves as a potential barrier and stops the inner electrons from leaving the surface of the metal. Thus, in free electron gas model, the movement of free electrons in a metal is equivalent to the movement of a free electron gas inside a 'potential energy box' which, in one-dimensional case, is represented by a line as shown in Fig. 5.1.

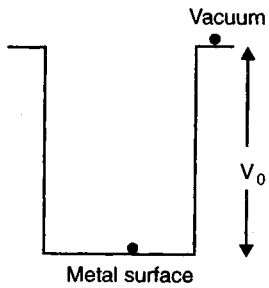


Fig. 5.1. *Metallic surface bounded by potential barrier V_0 which represents the difference in potential energy of a stationary electron present at the surface of the metal and just outside it (in vacuum).*

Based on the Drude's considerations that the electron gas behaves as a perfect gas, H.A. Lorentz postulated in 1909 that the electrons constituting the electron gas obey Maxwell-Boltzmann statistics under equilibrium conditions. The combined ideas of Drude and Lorentz constitue the *Drude-Lorentz theory.* As the theory is based on the classical ideas of Maxwell-Boltzmann statistics, it is called the classical theory.

The theory has been successfully applied to explain the various properties of metals. For example, it proves the validity of Ohm's law. The free electrons in a metal move in random directions and do not constitute a current until an electric field is applied across the metal which accelerates the electrons in a particular direction. The electrons, however, cannot be accelerated indefinitely. During their motion, the electrons suffer elastic collisions with the metal ions which slow down their speed. This gives rise to a steady state current of magnitude proportional to the voltage applied provided the temperature remains constant. This leads to Ohm's law. Also as the free electrons can move easily, the metals exhibit high electrical and thermal conductivities. Moreover, since the electrons move freely inside the metals irrespective of the crystal structure, the ratio of the electrical conductivity, σ, to the thermal conductivity, K, should be constant for all metals at a constant temperature, i.e.,

$$\frac{\sigma}{K} = \text{constant}$$

This is called the *Wiedemann-Franz law* and has been realized in practice. The theory also explains the high lustre and complete opacity of metals. The opacity is due to absorption of all the incident electromagnetic radiations by free electrons which are then set into forced oscillations. The electrons return to their normal states by emitting the same amount of energy in all directions, thus producing metallic lustre.

Besides these successes, the theory also met with a number of failures. It correctly predicted the room temperature resistivity of various metals but the temperature dependence of resistivity could not be established accurately. The theory predicted that resistivity varies as $\sqrt{T}$ whereas actually it is found to vary linearly with temperature. The theory failed to explain the heat capacity and paramagnetic susceptibility of the conduction electrons. These

properties are based on interactions of free electrons with the external sources of energy which may be thermal or magnetic in nature. The application of Maxwell-Boltzmann statistics to this theory allows all the free electrons to gain energy which results in much higher values of heat capacity and paramagnetic susceptibility. As we shall describe later, the use of Fermi-Dirac or quantum statistics allows only a fraction of the total number of free electrons to gain energy and the values of heat capacity and paramagnetic susceptibility thus obtained match with the observed values. These are the principal achievements of the Sommerfeld's theory. The classical theory is also unable to account for the occurrence of long mean free paths (~10^8 to 10^9 interatomic spacings, or more than one centimetre) at low temperatures.

5.2 SOMMERFELD'S QUANTUM THEORY

Considering the free nature of valence electrons as assumed in the classical theory, Sommerfeld treated the problem quantum mechanically using the Fermi-Dirac statistics rather than the classical Maxwell-Boltzmann statistics. The possible electronic energy states in the potential energy box and the distribution of electrons in these states are then determined using quantum statistics. We describe the one-dimensional and three-dimensional cases separately.

5.2.1 Free Electron Gas in One-Dimensional Box

Consider an electron of mass m which is bound to move in a one-dimensional crystal of length L. The electron is prevented from leaving the crystal by the presence of a large potential energy barrier at its surfaces. Although the barriers extend over a few atomic layers near the surface, these are taken infinitely large for the sake of simplicity. The problem is identical to that of an electron moving in a one-dimensional potential box which is represented by a line and is bounded by infinite potential energy barriers as shown in Fig. 5.2. The potential energy within the crystal or box is assumed to be zero. Thus, we have

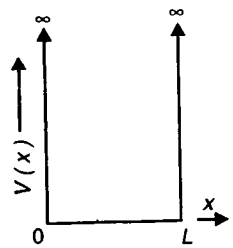


Fig. 5.2 One-dimensional potential box bounded by infinite potential energy barriers.

$$V(x) = 0 \quad \text{for } 0 < x < L$$
$$V(x) = \infty \quad \text{for } x \leq 0 \text{ and } x \geq L \qquad (5.1)$$

The wave function ψ_n of the electron occupying the nth state is determined from the solution of the Schrodinger equation, i.e.,

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2}(E_n - V)\,\psi_n = 0 \qquad (5.2)$$

where E_n represents the kinetic energy of the electron in the nth state and V is its potential energy. Since $V = 0$ inside the box, Eq. (5.2) becomes

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2} E_n\psi_n = 0 \tag{5.3}$$

The general solution to this equation is

$$\psi_n(x) = A \sin kx + B \cos kx \tag{5.4}$$

where A and B are arbitrary constants to be determined from the boundary conditions and k is given by

$$k = \sqrt{\frac{2mE_n}{\hbar^2}} \tag{5.5}$$

The boundary conditions are

$$\psi_n(0) = 0 \qquad \text{and} \qquad \psi_n(L) = 0 \tag{5.6}$$

These conditions are based on the fact that at $x = 0$ and L, $V \to \infty$ and the product $V(x)\,\psi_n(x)$ in Eq. (5.2) also approaches infinity. Thus in order that the wave function $\psi_n(x)$ may be continuous, the kinetic energy E_n must also become infinite which is not feasible. Hence $\psi_n(x)$ must vanish for $x = 0$ and L.

For $x = 0$, Eq. (5.4) gives B = 0 and the solution (5.4) becomes

$$\psi_n(x) = A \sin kx \tag{5.7}$$

Also, since $\psi_n(L) = 0$, Eq. (5.7) yields

$$\sin kL = 0$$

or

$$k = \frac{n\pi}{L} \tag{5.8}$$

where n = 1, 2, 3,....Thus the expression (5.7) for the allowed wave function becomes

$$\psi_n(x) = \text{A} \sin\left(\frac{n\pi}{L}\right)x \tag{5.9}$$

The allowed energy values can be obtained from Eqs. (5.5) and (5.8) as

$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2 = \frac{n^2h^2}{8mL^2} \tag{5.10}$$

or

$$E_n \propto n^2$$

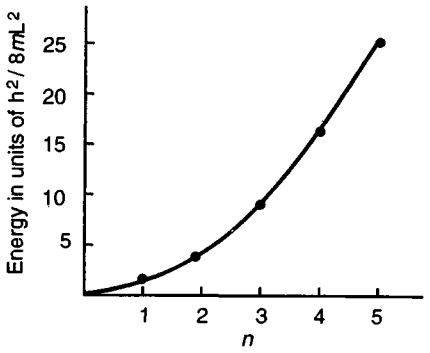


Fig. 5.3. E_n *versus n for a one-dimensional crystal.*

It is apparent from Eqs. (5.9) and (5.10) that the allowed wave functions $\psi_n(x)$ and the allowed energy values E_n exist only for integral values of n. The number n is called the quantum number. Hence the energy spectrum consists of discrete energy levels where the spacing between the levels is determined by the values of n and L. It decreases with increase in L. If L is of the order of a few centimetres, the energy levels form almost a continuum. But if L has atomic dimensions, the spacing between the levels becomes appreciable. The plot of E_n versus n is shown in Fig. 5.3.

The constant A in Eq. (5.9) is determined by using the condition that the probability of finding an electron somewhere on the line is unity, i.e.,

$$\int_0^L \psi_n^*(x)\psi_n(x)dx = 1$$

Using Eq. (5.9), we get

$$A^2 \int_0^L \sin^2\left(\frac{n\pi}{L}x\right)dx = 1$$

or
$$\frac{A^2}{2}\int_0^L\left[1-\cos\left(\frac{2n\pi}{L}x\right)\right]dx = 1$$

or
$$\frac{A^2}{2}\int_0^L dx = 1$$

or
$$A = \sqrt{\frac{2}{L}}$$

Substituting it in Eq. (5.9), we get the normalized wave function as

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right) \qquad (5.11)$$

The energy levels and the wave functions corresponding to n = 1, 2, 3 and 4 are shown in Fig. 5.4.

We now discuss the distribution of electrons among the allowed energy levels and determine the Fermi energy and the density of states.

(i) Fermi Energy

The electrons are distributed among the various possible energy levels in accordance with the Pauli's exclusion principle which states that no two electrons can have all their quantum numbers identical, i.e., each orbital or quantum state can be occupied by at the most one electron. But the different states may have different values (non-degenerate case) or the same value (degenerate case) of energy. In a solid, an electron in a conduction electronic state has the quantum numbers n and m_s where n is the principal quantum number and m_s is the magnetic spin quantum number. Each set of values of n and m_s define a quantum state. For each value of n, m_s can have two possible values, +1/2 or −1/2. This means that each energy level defined by the quantum number n can have two quantum states and hence can accommodate a maximum of two electrons, one with spin up and the other with spin down. In other words, each energy level is doubly degenerate. For example, if there are seven electrons of appropriate spins then, in the ground state, these can be accommodated in four energy levels; the first three levels would have two electrons each with opposite spins and the fourth level would contain the last unpaired electron. These four energy levels are represented by the values 1, 2, 3 and 4 of the quantum number n. Thus, we find that if the total number of electrons to be accommodated is seven, the energy levels with $n < 4$ would be occupied while the level with $n > 4$ would be empty. The topmost filled energy level at 0 K is known as the *Fermi level* and the energy corresponding to this level is called the *Fermi energy*, E_F.

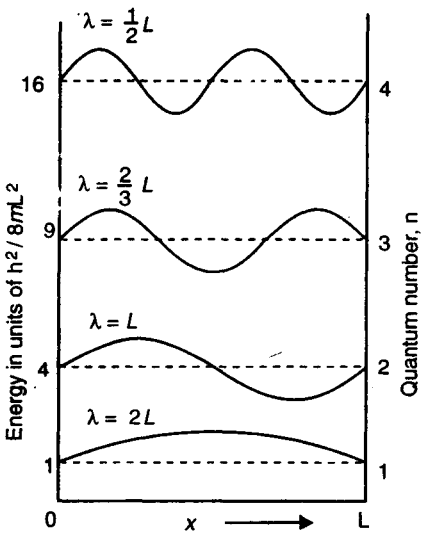


Fig. 5.4. *First four wave functions (solid lines) and the corresponding energy levels (broken lines) of an electron in a one-dimensional crystal.*

If N is the total number of electrons to be accommodated on the line then, for even n, we can write

$$2n_F = N \tag{5.12}$$

where n_F represents the principal quantum number of the Fermi level. Thus, for n = n_F, Eq. (5.10) gives

$$E_F = \frac{\hbar^2}{2m}\left(\frac{n_F\pi}{L}\right)^2 = \frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2 \tag{5.13}$$

Thus the value of the Fermi energy depends upon the length of the box the number of electrons in the box. For example, taking

$$\frac{N}{L} = 0.5 \text{ electrons/Å or } 5 \times 10^7 \text{ electrons/cm},$$

the Eq. (5.13) gives

$$E_F = 3.7 \times 10^{-12} \text{ erg} = 2.4 \text{ eV}$$

Thus if we accommodate 5×10^7 electrons on one centimetre length of the line, the energy of the topmost electron would be 2.4 eV.

(ii) Total Energy

The total energy, E_o, of all the N electrons in the ground state is determined by summing up the energies of the individual electrons. For N electrons, the number of filled energy levels is $N/2$ and E_0 is given by

$$E_0 = 2 \sum_{n=1}^{N/2} E_n$$

Here the factor of 2 appears because each level contains two electrons with equal energies. Using Eq. (5.10), we get

$$E_0 = 2\,\frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 \sum_{n=1}^{N/2} n^2 \tag{5.14}$$

Since

$$\sum_{n=1}^{s} n^2 = \frac{1}{6}s\left(2s^2+3s+1\right)$$

$$\cong \frac{1}{3}s^3 \qquad \text{for} \quad s \geq 1$$

$$\therefore \qquad \sum_{n=1}^{N/2} n^2 \cong \frac{1}{3}\left(\frac{N}{2}\right)^3$$

Hence Eq. (5.14) becomes

$$E_0 = 2\,\frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 \frac{1}{3}\left(\frac{N}{2}\right)^3 = \frac{1}{3}\,\frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2 N$$

or, using Eq. (5.13), we obtain

$$E_0 = \frac{1}{3}NE_F \tag{5.15}$$

Thus, for one-dimensional crystal, the average kinetic energy in the ground state is one-third of the Fermi energy.

(iii) Density of States

The density of states is defined as the number of electronic states present in a unit energy range. It is denoted by $D(E)$ and is given by

$$D(E) = \frac{dn}{dE} \tag{5.16}$$

where dn represents the number of electronic quantum states present in the energy interval E and $E + dE$. For a free electron gas, since each energy level contains two electronic states, one with spin up and the other with spin down, the actual density of states is twice the value given by Eq. (5.16). Thus, we have

$$D(E) = 2\frac{dn}{dE} \tag{5.17}$$

From. Eq. (5.10), we obtain

$$\frac{dE}{dn} = \frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 2n = \frac{h^2 n}{4mL^2}$$

Therefore, Eq. (5.17) becomes

$$D(E) = 2\left(\frac{4mL^2}{nh^2}\right) = \frac{8mL^2}{h^2}\,\frac{1}{n}$$

Again, from Eq. (5.10), we get

$$\frac{1}{n} = \left(\frac{h^2}{8mL^2 E}\right)^{1/2}$$

Hence

$$D(E) = \left(\frac{8mL^2}{h^2 E}\right)^{1/2} = \frac{4L}{h}\left(\frac{m}{2E}\right)^{1/2} \tag{5.18}$$

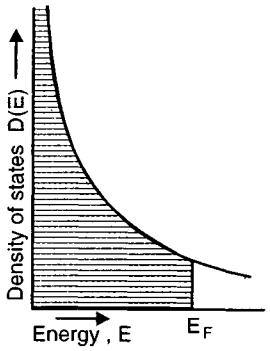


Fig. 5.5. *Variation of density of electronic states with energy for a one-dimensional metallic crystal. At* 0 K, *all the states upto the Fermi level are filled.*

The plot of $D(E)$ versus E is shown in Fig. 5.5. It indicates that all the levels present below the Fermi level are filled and all those present above it are empty. This type of situation, in fact, exists at absolute zero. Thus at 0 K, the Fermi level divides the filled and unfilled levels in the metallic crystals.

5.2.2 Free Electron Gas in Three Dimensions

Consider now the behaviour of the free electron gas in a three-dimensional metallic crystal. The potential of the electrons inside the crystal is constant and may be taken as zero, whereas it has a large value outside the crystal. For simplicity, the three-dimensional crystal may be regarded as a cubical box having length of the edge equal to L. The free particle Schrodinger equation in three dimensions is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi_k(\mathbf{r}) = E_k\psi_k(\mathbf{r})$$

or
$$\nabla^2\psi_k(\mathbf{r}) + \frac{2m}{\hbar^2}E_k\psi_k(\mathbf{r}) = 0 \qquad (5.19)$$

where
$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$$

is the Laplacian operator and E_k is the total energy (kinetic energy in the present case) of the electron in the k-state. Since the electrons are confined to a cubical box of edge L, the solution to Eq. (5.19) is just an extension of the one-dimensional normalized wave function (5.11), i.e.,

$$\psi_k(\mathbf{r}) = \sqrt{\frac{8}{L^3}}\sin\left(\frac{\pi n_x x}{L}\right)\sin\left(\frac{\pi n_y y}{L}\right)\sin\left(\frac{\pi n_z z}{L}\right) \qquad (5.20)$$

where n_x, n_y and n_z are positive integers and $\sqrt{8/L^3}$ is the normalising constant. This equation represents a standing wave solution. It is, however, more convenient to work with the plane travelling wave solution of the type

$$\psi_k(\mathbf{r}) = A\, e^{i\mathbf{k}.\mathbf{r}} = Ae^{i\left(k_x x + k_y y + k_z z\right)} \qquad (5.21)$$

where $$k^2 = \frac{2m}{\hbar^2} E_k \tag{5.22}$$

and A is an arbitrary constant. Such wave functions must satisfy the periodic boundary conditions, i.e., they must be periodic in x, y and z with period equal to L. These boundary conditions are

$$\begin{aligned} \psi(x + L, y, z) &= \psi(x, y, z) \\ \psi(x, y + L, z) &= \psi(x, y, z) \\ \psi(x, y, z + L) &= \psi(x, y, z) \end{aligned} \tag{5.23}$$

An application of the first boundary condition in (5.23) to the wave function (5.21) yields

$$\exp[i\{k_x(x + L) + k_y y + k_z z\}] = \exp\,[i\{k_x x + k_y y + k_z z\}]$$

or

$$\exp(ik_x L) = 1$$

or

$$k_x = 0, \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \text{------} \pm \frac{2n\pi}{L} \tag{5.24}$$

Similar results are obtained for k_y and k_z. This implies that any component of k is of the form $\frac{2n\pi}{L}$, where n is a positive or negative integer. These three components of k also form the quantum numbers of the problem in addition to the quantum number m_s which represents the spin direction. Thus the state of an electron is specified completely by a set of four quantum numbers k_x, k_y, k_z and m_s.

The allowed eigen values of the state or orbital with wave vector **k** are obtained from Eq. (5.22) as

$$E_k = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right) \tag{5.25}$$

where the magnitude of the wave vector **k** is related to the wavelength λ as

$$k = \frac{2\pi}{\lambda} \tag{5.26}$$

Thus it is easy to see that the energy spectrum consists of discrete energy levels. These levels are generally very close to each other (energy difference ~ 10^{-15} eV) and may be regarded as continuous for most of the purposes.

These levels are also said to be *quasi-continuous*. In fact, the spacing between the levels depends upon the dimensions of the box. For dimensions of the order of a centimetre, the levels are so closely spaced that they almost form a continuum as predicted by the classical mechanics. If, however, the box has atomic dimensions, the energy levels are widely spaced and discrete.

The constant A in Eq. (5.21) can be determined by using the normalization condition, i.e.,

$$\int_0^V \psi^*(\mathbf{r})\,\psi(\mathbf{r})\,dV = 1$$

or
$$\int_0^L\int_0^L\int_0^L A^2 e^{-i\mathbf{k}.\mathbf{r}} e^{i\mathbf{k}.\mathbf{r}}\,dx\,dy\,dz = 1$$

or
$$A = \left(\frac{1}{L^3}\right)^{1/2} = \left(\frac{1}{V}\right)^{1/2}$$

Hence the normalized wave function is written as

$$\psi_k(\mathbf{r}) = \left(\frac{1}{V}\right)^{1/2} e^{i\mathbf{k}.\mathbf{r}} \tag{5.27}$$

The distribution of electrons among the allowed energy levels and the density of states are described in the following sections.

(i) Filling of Energy Levels

As described earlier, the distribution of electrons in the allowed energy levels follows the Pauli's exclusion principle. In the present case, we have a set of four quantum numbers k_x, k_y, k_z and m_s which represents a state of the system. Apparently, each state can be occupied by at the most one electron. It also follows from the expression (5.25) that a set of values of k_x, k_y and k_z defines an energy level. Now since for each set of k_x, k_y and k_z the fourth quantum number can have two values $\pm 1/2$, it follows that each energy level contains two quantum states of orbitals and hence can accommodate a maximum of two electrons, one with spin up and the other with spin down. In other words, each energy level is doubly degenerate. Thus a total of N non-interacting electrons at 0 K can be filled in $N/2$ energy levels. The topmost filled level is the $(N/2)$th level and all the levels lying

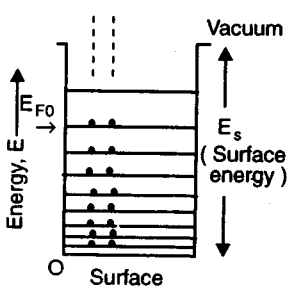


Fig. 5.6. *Sommerfeld's free electron model at* 0 K. *All the levels below* E_{F_0} *are filled and all those above* E_{F_0} *are empty.*

above it are empty. This level is, therefore, the *Fermi level* as it divides the filled and empty levels at 0 K. The energy of the Fermi level, or the Fermi energy, is denoted by E_{F0}. The distribution of electrons among the energy levels present inside the box at 0 K is shown in Fig. 5.6. Thus we note that, unlike the classical theory, the Sommerfeld's quantum theory does not allow the condensation of all the electrons into the state of zero energy even at absolute zero. These electrons are rather distributed among the discrete energy levels having energies ranging from 0 to E_{F0}.

We now discuss the effect of temperature on the occupancy of energy levels. It is apparent that, for temperatures greater than 0 K, the Fermi level may not be the topmost filled level since some of the electrons from the filled energy levels may be excited to the higher levels. Thus some of the levels below E_{F0} would be empty while some above it would be occupied. The probability that a particular quantum state of energy E is occupied at a temperature T is given by the so called *Fermi function*

$$f(E) = \frac{1}{\exp\left(\frac{E - E_F}{k_B T}\right) + 1} \tag{5.28}$$

where E_F is the Fermi energy. The plot of $f(E)$ versus E is shown in Fig. 5.7 for different temperatures.

At absolute zero, the Eq. (5.28) gives

$$\begin{aligned} f(E) &= 1 \quad \text{for} \quad E \le E_{F0} \\ &= 0 \quad \text{for} \quad E > E_{F0} \end{aligned} \tag{5.29}$$

This indicates that all the energy states below E_{F0} are occupied and all the states above it are empty, i.e., the Fermi distribution function is a step function. Thus, at 0 K, the Fermi level E_{F0} represents the highest filled energy level.

For temperatures greater than 0 K but less than the melting point of the metal such that $k_B T << E_F$, the distribution function loses its step character. The probability of occupation, $f(E)$, decreases gradually from 1 to 0 near E_F. This indicates that some of the states below E_F are empty while some others above it are filled. This is because some

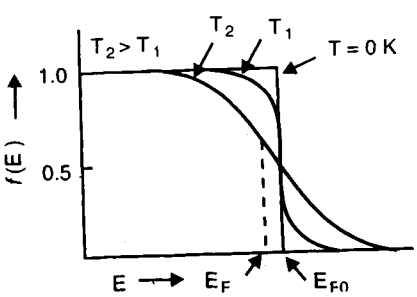


Fig. 5.7. *Schematic representation of the Fermi distribution function for three different temperatures. The variation of the Fermi energy with temperature is also shown.*

of the electrons from the energy states below E_F gain thermal energy and get excited to the states above E_F. At $E = E_F$, Eq. (5.28) gives

$$f(E_F) = \frac{1}{2}$$

Thus, for temperatures greater than 0 K, the Fermi level may be defined as the level where the probability of occupation is 1/2. Unlike E_{Fo}, it is not the topmost filled level; instead, it lies between the filled levels and empty levels. Apparently, the position of the Fermi level is not fixed, but changes with temperature. An approximate relationship between E_F and E_{Fo} is given as follows :

$$E_F \cong E_{Fo}\left[1-\frac{\pi^2}{12}\left(\frac{k_BT}{E_{F_o}}\right)^2\right] \tag{5.30}$$

However, in case of metals, since the spacing between the levels is quite small ($\sim 10^{-19}$ eV), the highest filled energy level is usually taken as the Fermi level.

For energies below E_F such that $(E_F - E) >> k_BT^*$, $f(E) \cong 1$, provided T does not differ much from 0 K. Thus it is only in the vicinity of E_F minus a few k_BT that $f(E)$ becomes less than unity, i.e., some of the states below E_F would be empty and some above E_F would be filled. It, therefore, follows that as the temperature is raised above 0 K, all the electrons would not gain energy as expected classically, but only those which lie within an energy range k_BT below the Fermi level can do so. This is because the electrons present near the Fermi level can jump to the vacant higher energy states after acquiring energy of the order of k_BT, but the electrons present well below the Fermi level cannot do so due to the non-availability of empty states within the range k_BT. Thus, according to quantum mechanics, only a small fraction of the electrons can gain thermal energy and get excited to the energy states. The value of this fraction is $\frac{k_BT}{E_F} \cong 0.01$ at room temperature for $E_F = 3.0$ eV.

(ii) Density of Available Electron States, D(E)

As defined earlier, the density of states, $D(E)$, is the total number of available electronic states (or orbitals) per unit energy range at energy E. To obtain the expression for $D(E)$, we consider the linear momentum **p** which, in quantum mechanics, is represented by the operator

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* According to the classical mechanics, k_BT represents the energy acquired by a particle at a temperature T K. At room temperature, $k_BT \sim 0.025$ eV.

$$\mathbf{p} = -i\hbar\nabla$$

When operated on Eq. (5.21), it gives

$$\mathbf{p}\psi_k(\mathbf{r}) = -i\hbar\nabla\psi_k(\mathbf{r}) = \hbar\mathbf{k}\psi_k(\mathbf{r}) \tag{5.31}$$

This indicates that the plane wave function ψ_k is an eigen function of the linear momentum with the eigen value $\hbar\mathbf{k}$. The particle velocity is given by

$$v = \frac{\hbar k}{m} \tag{5.32}$$

In a system of N free electrons, the occupied states or orbitals in the ground state may be represented by points inside a sphere in the **k**-space. The energy corresponding to the surface of the sphere then represents the Fermi energy, Let $\mathbf{k_F}$ be the wave vector from the origin of the **k**-space to the surface of the sphere as shown in Fig. 5.8. Then, using Eq. (5.25), the Fermi energy is written as

$$E_F = \frac{\hbar^2}{2m} k_F^2 \tag{5.33}$$

It is observed from Eqs. (5.24) and Fig. 5.8 that there is one allowed wave vector or one distinct triplet of quantum numbers k_x, k_y and k_z which corresponds to the volume element $(2\pi/L)^3$ of **k**-space. Thus, in the sphere called the *Fermi sphere* of volume $(4\pi/3)k^3_F$, the total number of electronic states or orbitals is

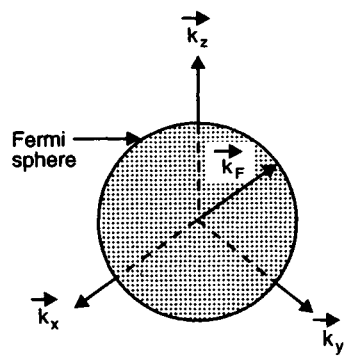


Fig. 5.8. *A sphere of radius k_F in the* **k***-space containing points which represent the occupied states of a system of N free electrons in the ground state.*

$$2\,\frac{\frac{4\pi}{3}k_F^3}{\left(\frac{2\pi}{L}\right)^3} = \frac{L^3}{3\pi^2}k_F^3 = \frac{V}{3\pi^2}k_F^3 = N \tag{5.34}$$

where the factor 2 appears because there are two allowed values of m_s, and hence two electronic states, corresponding to each value of k. The number of states has been set equal to the number of electrons, N. From Eq. (5.34), we obtain

$$k_F = \left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.35}$$

This shows that the value of k_F depends on the concentration of electrons, N/V, and is independent of the mass of the electron. The expression for the Fermi energy is obtained by using Eq. (5.35) in (5.33) as

$$E_F = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} \tag{5.36}$$

i.e., E_F depends on both the electronic concentration and mass. The total number of electrons is, therefore, given by

$$N = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2} \tag{5.37}$$

The electron velocity, v_F, at the Fermi surface is obtained from Eqs. (5.32) and (5.35) as

$$v_F = \frac{\hbar k_F}{m} = \frac{\hbar}{m}\left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.38}$$

The density of states function, $D(E)$, is obtained by using the fact that, in the ground state of the system, all the energy states below E_F are occupied and the total number of states is equal to the total number of electrons, i.e.,

$$\int_0^{E_F} D(E)dE = N \tag{5.39}$$

Substituting the value of N from Eq. (5.37), we get

$$\int_0^{E_F} D(E)dE = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2}$$

Expressing the integral in an indefinite form, we obtain

$$\int D(E)dE = \frac{V}{3\pi^2}\left(\frac{2mE}{\hbar^2}\right)^{3/2} = \int \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}\,dE$$

or

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2} \tag{5.40}$$

The variation of $D(E)$ with E is parabolic and is shown in Fig. 5.9. The value of $D(E)$ also increases with increase in crystal volume; this is to accommodate the total number of electrons N which also increases [Eq. (5.37)] with volume of the crystal. At the Fermi surface, the density of states is obtained by putting $E = E_F$ in Eq. (5.40), i.e.,

$$D(E_F) \equiv \frac{dN}{dE_F} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{1/2} \tag{5.41}$$

This result may also be obtained and expressed in a simple way by taking the natural logarithm of the expression (5.37), i.e.,

$$\ln N = \frac{3}{2}\ln E_F + \text{constant}$$

or

$$\frac{dN}{N} = \frac{3}{2}\frac{dE_F}{E_F} \tag{5.42}$$

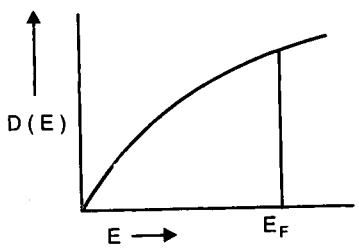


Fig. 5.9. *Variation of density of electronic states with energy for a free electron gas in three dimensions.*

Thus, within a factor of the order of unity, the number of electronic states or orbitals per unit energy range at the Fermi energy is equal to the total number of conduction electrons divided by the Fermi energy.

The density of filled electronic states, $N(E)$, at any temperature T is obtained by multplying the density of states, $D(E)$, at 0 K by the probability of occupation, $f(E)$, of the quantum state E at that temperature, i.e.,

$$N(E) = D(E)\, f(E) \tag{5.43}$$

The function $f(E)$ is the Fermi function and is defined by Eq. (5.28). The number of filled electronic states in the energy range E and $E + dE$ at temperature T is then given by

$$N(E)dE = D(E)\, f(E)dE \tag{5.44}$$

We now discuss some parameters of free electron gas at absolute zero.

(a) The Fermi Energy, E_{F_0}

The total number of electrons can be obtained by integrating Eq. (5.43), i.e.,

$$N = \int_0^\infty N(E)dE = \int_0^\infty D(E)f(E)dE \tag{5.45}$$

Using the expression (5.40), we get

$$N = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^\infty E^{1/2} f(E)dE$$

$$= \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\left[\int_0^{E_{F0}} E^{1/2} f(E)dE + \int_{E_{F0}}^\infty E^{1/2} f(E)dE\right.$$

Now using Eqs. (5.29), we get

$$N = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^{E_{Fo}} E^{1/2}dE$$

$$= \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_{Fo}^{3/2}$$

or
$$E_{Fo} = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} \tag{5.46}$$

Putting $N/V = n$, the electron concentration, we obtain

$$E_{Fo} = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3} = \frac{h^2}{8m}\left(\frac{3n}{\pi}\right)^{2/3} \tag{5.47}$$

This shows that the Fermi energy at 0 K can be calculated simply from the electron concentration n. We also conclude from the above relationship that the electrons have some definite energy even at absolute zero. This is something entirely different from the classical theory according to which the electrons should have zero energy at absolute zero.

(b) Average Kinetic Energy, $\bar{E}_0$

The average kinetic energy of an electron at absolute zero is calculated as

$$\bar{E}_0 = \frac{1}{N}\int_0^\infty ED(E)f(E)dE$$

Using the expression (5.40) for $D(E)$, we obtain

$$\bar{E}_o = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{N}\left[\int_0^{E_{Fo}} EE^{1/2}f(E)dE + \int_{E_{Fo}}^\infty EE^{1/2}f(E)dE\right]$$

Now using Eq. (5.29), we get

$$\bar{E}_0 = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{N}\int_0^{E_{Fo}} E^{3/2}dE$$

$$= \frac{2}{5}\frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{N}E_{Fo}^{5/2} \tag{5.48}$$

Substituting the value of N from Eq. (5.37), we obtain

$$\bar{E}_0 = \frac{3}{5}E_{Fo} \tag{5.49}$$

This relates the average kinetic energy of an electron at absolute zero to the Fermi energy at the same temperature.

5.3 APPLICATIONS OF THE FREE ELECTRON GAS MODEL

The electronic properties of metals can be broadly divided into the following two groups :

(i) Static properties

(ii) Transport properties

The static properties are those which can be treated effectively by considering just the energy levels or the distribution of energy levels to which the electrons belong. More specifically, we need to consider the overall change in potential of the electrons without investigating deeply the detailed process producing the transitions. These properties include various electron emission properties, the magnetic properties and the properties like heat capacity and contact potential. These properties arise as a result of the excitations of the electrons by light, thermal energy, strong electric fields, etc.

The transport properties are those which can be treated by considering the detailed response of the electrons to an external field, i.e., by considering the acceleration properties of the electrons. These properties include electrical and thermal conductivities, Hall effect, thermoelectricity, etc.

We consider below the electronic specific heat of metals.

5.3.1 Electronic Specific Heat

The average kinetic energy of a free electron as given by the classical statistical mechanics is

$$\bar{E}_0 = \frac{3}{2}k_BT$$

If the metal contains N free electrons, then the total kinetic energy becomes

$$E = N\bar{E}_0 = \frac{3}{2} Nk_BT$$

The electronic specific heat is, therefore, given by

$$C_v = \frac{3}{2} Nk_B \tag{5.50}$$

The measurement of optical reflection coefficient of metals indicates that the number of electrons per atom in metals is of the order of unity. Therefore, considering one gram atom of the metal, the value of electronic specific heat is obtained as (3/2)R or 3 cal $g^{-1}K^{-1}$ approximately. On the other hand, the specific heat associated with the lattice vibrations at high temperatures is about 3R or 6 cal $g^{-1}K^{-1}$. Thus one might conclude that the specific heat of metals should be about 50% greater than the specific heat of insulators. This is contradictory to the experimental observations which indicate that the electronic contribution to the specific heat is very small. At room termperature, the electronic contribution is not more than 0.01 of the value given by Eq. (5.50). Also, this contribution decreases linearly to zero as T approaches zero. It, therefore, appears that all the electrons might not contribute to the specific heat; instead, only a small fraction of the total number of electrons might contribute. This view is supported by the quantum theory.

As described earlier, the quantum mechanics suggests that only those electrons contribute to the specific heat which lie within an energy range k_BT below the Fermi level. This happens because when the electron gas is heated up to a temperature T, these electrons acquire energy of the order of k_BT and jump to the empty higher energy states. The deep lying electrons cannot do so because the unfilled energy states are not available to these electrons for excitation. These electrons, as such, do not contribute to the specific heat. The number of electrons which contribute to the specific heat is of the order $N\left(\frac{k_BT}{E_F}\right)$ or $N\left(\frac{T}{T_F}\right)$ where T_F is called the *Fermi termperature* and is defined by the equation

$$E_F = k_BT_F$$

Using the effective value of N, Eq. (5.50) becomes

$$C_v \cong \frac{3}{2} Nk_B\left(\frac{T}{T_F}\right) \tag{5.51}$$

This indicates that C_v is proportional to T and approaches zero as $T \to 0$. Using typical values of T and T_F as 300 K and 3000 K respectively, the above equation gives the value of C_v as

$$C_v \cong (0.01)\left(\frac{3}{2}\right)Nk_B$$

The specific heat calculated from this equation agrees with the experimental observations. Thus the quantum mechanics modifies the thermal behaviour of free electrons in a simple and satisfying manner.

SOLVED EXAMPLES

Example 5.1. The atomic radius of sodium is 1.86 Å. Calculate the Fermi energy of sodium at absolute zero.

Solution. The Fermi energy at 0 K is given by the relation

$$E_{Fo} = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3}$$

where N is the number of free (valence) electrons present in volume V of the metal. We consider a unit cell of sodium which is *bcc*.

Radius of the sodium atom, $r = 1.86 \times 10^{-10}$m

Volume of the unit cell, $V = a^3 = \left(\frac{4r}{\sqrt{3}}\right)^3$

$$= \left(\frac{4 \times 1.86 \times 10^{-10}}{\sqrt{3}}\right)^3$$

$$= 7.93 \times 10^{-29}\ \text{m}^3$$

Since an atom of sodium has only one valence electron, the number of free electrons in a unit cell of sodium is 2.

$$\therefore \quad \frac{N}{V} = \frac{2}{7.93 \times 10^{-29}} = 2.52 \times 10^{28}\ \text{electrons/m}^3.$$

$$\therefore \quad E_{Fo} = \frac{\left(1.05 \times 10^{-34}\right)^2}{2 \times 9.1 \times 10^{-31}}\left(3\pi^2 \times 2.52 \times 10^{28}\right)^{2/3}$$

$$= 4.98 \times 10^{-19} \text{ J}$$

$$= 3.11 \text{ eV}.$$

Example 5.2. Derive pressure versus volume relationship for a free electron gas at 0 K. Hence obtain an expression for the bulk modulus in terms of the average kinetic energy.

Solution. From thermodynamics, we have

$$P = -\frac{\partial E}{\partial V}$$

where E is the internal energy of a system of particles occupying a volume V at pressure P. For a free electron gas containing N electrons with average kinetic energy $\bar{E}_0$ at 0 K, the energy E may be replaced by $N\bar{E}_0$. Therefore, we have

$$P = -N \frac{\partial \bar{E}_0}{\partial V}$$

Using Eq. (5.49), we obtain

$$P = -\frac{3}{5} N \frac{\partial E_{F0}}{\partial V}$$

Now, from Eq. (5.46), we have

$$E_{F0} = \frac{\hbar^2}{2m} \left(\frac{3\pi^2 N}{V} \right)^{2/3}$$

which gives

$$\frac{\partial E_{F0}}{\partial V} = -\frac{2}{3} \frac{\hbar^2}{2m} \left(3\pi^2 N\right)^{2/3} \left(\frac{1}{V}\right)^{5/3}$$

$$\therefore \quad P = \frac{2}{5} N \frac{\hbar^2}{2m} \left(3\pi^2 N\right)^{2/3} \left(\frac{1}{V}\right)^{5/3} \tag{5.52}$$

or

$$P = \frac{2}{5} \frac{NE_{F0}}{V} \tag{5.53}$$

This is the pressure versus volume relationship for a free electron gas at 0 K. The bulk modulus is given by the expression

$$B = -V \frac{\partial P}{\partial V}$$

Differentiating (5.52) with respect to V, we get

$$\frac{\partial P}{\partial V} = -\frac{5}{3}\left[\frac{2}{5}\frac{N\hbar^2}{2m}\left(3\pi^2 N\right)^{2/3}\right]\left(\frac{1}{V}\right)^{7/3}$$

$$= -\frac{5}{3}\frac{P}{V}$$

$$\therefore \qquad B = \frac{5}{3}P = \frac{2}{3}\frac{NE_{Fo}}{V} = \frac{10}{9}\frac{N\bar{E}_o}{V} = \frac{10}{9}\frac{E}{V} \qquad (5.54)$$

where we have used Eqs. (5.53) and (5.49).

SUMMARY

1. According to the free electron gas model, the valence electrons of the atoms present in a metal move freely among the positive ion cores just like the molecules of an ideal gas. These electrons are also called the conduction electrons and are responsible for conduction of electricity.

2. The energy spectrum of a free electron gas present in one or three-dimensional box is discrete particularly when the box has atomic dimensions. For laboratory dimensions, the levels almost form a continuum. The allowed eigen values of the states with wave vector **k** are given as

$$E_k = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right)$$

where m is the mass of the electron.

3. The Fermi level is the level which divides the filled and unfilled states. It is the topmost filled level at 0 K. For T > 0 K, the probability of occupation of this level is 1/2. The position of the Fermi level changes with temperature and is expressed by the relation

$$E_F \cong E_{Fo}\left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{E_{Fo}}\right)^2\right]$$

where E_{Fo} is the Fermi energy at 0 K.

4. The Fermi energy varies with electron concentration n as

$$E_F = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}$$

5. The number of electronic states present in a unit energy range is called the density of states. In a three-dimensional box of volume V, the density of states at energy E is given by

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$$

At the Fermi level, $D(E)$ may be expressed as

$$D(E_F) = \frac{3}{2}\frac{N}{E_F}$$

where N is the total number of electrons.

6. The average kinetic energy of an electron at 0 K is related to the Fermi energy at the same temperature as

$$\bar{E}_0 = \frac{3}{5} E_{Fo}$$

7. The free electron gas model explains the electronic specific heat of metals by restricting the number of electrons contributing to specific heat to $N\left(\frac{k_B T}{E_F}\right)$. It yielded the expression for specific heat as

$$C_v \cong \frac{3}{2} N k_B \left(\frac{k_B T}{E_F}\right)$$

VERY SHORT QUESTIONS

1. What do you understand by free electron gas ?
2. What is Wiedemann-Franz law ?
3. Define the Fermi energy for metals. How does it depend on mass of the sample?
4. What is the significance of the Fermi distribution function?
5. Define density of states.
6. What do you understand by a Fermi gas ?
7. What is the Fermi sphere ?
8. Can a metal be associated with two Fermi temperatures ?

SHORT QUESTIONS

1. How does a free electron gas differ from an ordinary gas ?
2. How does the classical free electron theory lead to the Ohm's law?
3. Explain complete opacity and high lustre on the basis of the Drude-Lorentz free electron theory.
4. What are the failures of the Drude-Lorentz free electron theory ?
5. Obtain an expression for the energy eigen values of an electron using the Sommerfeld's free electron theory in one dimension.
6. Derive the three-dimensional wave function for the free electron gas and obtain its eigen values.
7. Write the expression for the Fermi-Dirac distribution function and discuss its behaviour with change in temperature.

8. How does the Fermi energy of a metal depend on temperature, concentration of electrons and the total number of electrons present in the metal ?
9. How does the quantum free electron theory explain the observed small values of electronic specific heat of metals ?

LONG QUESTIONS

1. Describe the free electron gas model of metals. How does it help to explain the lattice heat capacity of metals.
2. Obtain expressions for the Fermi energy, the total energy and the density of states for a free electron gas in one dimension. Show the variation of density of states with energy.
3. What is the Fermi energy? Calculate its value for the free electron gas at 0K and mention its significance.
4. Derive expressions for the Fermi energy and density of states for a free elctron gas in one dimension.
5. Show that for a one-dimensional metallic crystal, the average kinetic energy of electron in the ground state is one-third of the Fermi energy.
6. Show that the average kinetic energy per electron for a three-dimensional free electron gas at 0K is

$$\overline{E}_0 = (3/5)E_{F0}$$

where E_{F0} is the Fermi energy at 0K.

7. Discuss qualitatively why the electronic specific heat is temperature dependent and is much less than that expected from the classical behaviour of free electron gas.

PROBLEMS

1. Calculate the Fermi energy of electrons at 0K for a metal with electron density of $10^{28} m^{-3}$. (1.69eV)
2. Silver (*fcc*) has an atomic radius of 1.44 Å. Assuming silver to be monovalent metal, calculate the value of the Fermi energy, the Fermi temperature and the Fermi velocity.
   (5.50 eV, 63850K, 1.39×10^6 ms^{-1})
3. Aluminium metal crystallises in *fcc* structure. If each atom contributes single electron as free electron and the lattice constant is 4Å, calculate (treating conduction electrons as free electron Fermi gas)
   (*i*) Fermi energy (E_F) and the Fermi wave vector (k_F),
   (*ii*) Total kinetic energy of free electron gas per unit volume at 0K.
   Given $\hbar = 1.054 \times 10^{-27}$ erg-s; Electron rest mass = 9.11×10^{-28}g.
   (2.28 eV, 7.73×10^9 m^{-1}, 2.14×10^{28} eVm^{-3})
4. The density of potassium is 860 kgm^{-3}. Calculate the Fermi energy and compressibility of potassium at 0K. (Atomic mass of K = 39.1 amu) (2.02 eV, 3.51×10^{-10} $N^{-1}m^2$)
5. If Fermi temperature is of the order of 5×10^4K, provide a rough estimate of electronic specific heat at 100K. (» 0.025J)

CHAPTER -VI

BAND THEORY OF SOLIDS

6.1 INTRODUCTION

The free electron theory described in the previous chapter was successful in explaining the various electronic and thermal properties of metals, such as specific heat, paramagnetism, etc. However, there are various other properties which could not be explained by this theory. For example, the theory could not explain why certain solids have a large number of free electrons and thus behave as good conductors while certain others have hardly any electrons and are, therefore, insulators. It also could not account for the variation of resistivity with temperature for the latter type of materials. Furthermore, the properties of semiconductors could not be explained on the basis of this theory.

The failure of the free electron model is due to the oversimplified assumption that a conduction electron in a metal experiences a constant or zero potential due to the ion cores and hence is free to move about in a crystal; the motion being restrained only by the surface of the crystal. In fact, the potential due to ion cores is not constant and may change with position of the electron in the crystal. Some contribution to potential may also arise because of the other electrons present in the crystal. Thus the actual nature of potential under which an electron moves in a crystal is very complicated. To a reasonable approximation, the ion cores may be considered at rest and the potential experienced by an electron in a crystal is assumed to be periodic with period equal to the lattice constant as shown in Fig. 6.1 for a one-dimensional case. This assumption is based on the fact that the ion cores in a crystal are distributed periodically on the lattice sites. The potential contribution due to all other free electrons may be taken as constant. This type of periodic potential extends up to infinity in all directions except at the surface of the crystal where, due to interrruption in periodicity of the lattice, it has a shape similar to as shown on the right hand edge of the figure. This type of potential irregularity may, however, be ignored.

The periodic potential described above forms the basis of the band theory of solids. The behaviour of an electron in this potential is described by constructing the electron wave functions using one-electron approxima-

tion. In this approximation, the total wave function for the system is obtained from a combination of wave functions each one of which involves the coordinates of one electron. In other words, the field experienced by a particular electron is assumed to be the resultant of the field due to fixed nuclei and the average field due to the charge distribution of all other electrons. As we shall discuss later, the motion of an electron in a periodic potential yields the following results :

(a) There exist allowed energy bands separated by forbidden regions or band gaps.

(b) The electronic energy functions E(**k**) are periodic in the wave vector **k**.

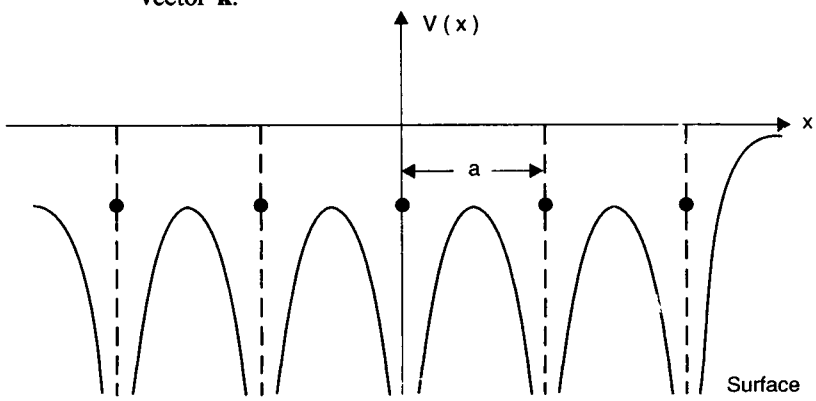


Fig. 6.1. *One-dimensional representation of potential experienced by an electron in a perfectly periodic crystal lattice with lattice constant a. The potential near the surface is shown on the right.*

These results are in contrast with those obtained from the free electron theory (constant or zero potential case) in the sense that, in the free electron theory, E varies with k as

$$E = \frac{\hbar^2 k^2}{2m}$$

i.e., there is no upper limit to the energy and k can have discrete values, which means that the energy levels are discrete and may have any spacing depending on the dimensions of the box. The existence of band gaps is the most important new property which emerges when the free electron model is extended to include the effect of periodic potential of the ion cores. It will be shown that these band gaps result from the interaction of the conduction electron waves with ion cores of the crystal and are of decisive significance to determine whether a solid is a conductor, a semiconductor, or an insulator. Besides this, we shall also encounter some other remarkable properties of electrons in crystals. We first determine the electron wave functions for a periodic lattice.

6.2 THE BLOCH THEOREM

As described in case of the free electron theory, the one-dimensional Schrodinger equation for an electron moving in a constant potential V_o is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_o)\psi = 0 \tag{6.1}$$

The solutions to this equation are plane waves of the type

$$\psi(x) = e^{\pm ikx} \tag{6.2}$$

where

$$(E - V_o) = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m} = E_{kin}$$

For an electron moving in a one-dimensional periodic potential, $V(x)$, the Schrodinger equation is written as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi = 0 \tag{6.3}$$

Since the potential is periodic with period equal to the lattice constant, a, we have

$$V(x) = V(x + a)$$

There is an important theorem known as the *Bloch theorem* or the *Floquet's theorem* concerned with the solutions to Eq. (6.3). According to this theorem, the solutions to Eq. (6.3) are plane waves of the type of Eq. (6.2) which are modulated by a function $u_k(x)$ having the same periodicity as that of the lattice. Thus the solutions are of the form

$$\psi(x) = e^{\pm ikx}\, u_k(x) \tag{6.4}$$

with

$$u_k(x) = u_k(x + a) \tag{6.5}$$

The wave functions of the type of Eq. (6.4) are called the *Bloch functions.*

Proof of the Bloch Theorem

Let $g(x)$ and $f(x)$ be two real and independent solutions to the second order differential equation (6.3). Then the general solution can be written as

$$\psi(x) = A\, f(x) + B\, g(x) \tag{6.6}$$

where A and B are arbitrary constants. Since $V(x) = V(x + a)$, the functions $g(x + a)$ and $f(x + a)$ must also be the solutions to Eq. (6.3) in addition to $g(x)$ and $f(x)$

However, Eq. (6.3) cannot have more than two independent solutions, the functions $g(x+a)$ and $f(x+a)$ must be expressible in terms of the functions $g(x)$ and $f(x)$, i.e.,

$$\left.\begin{aligned} f(x+a) &= \alpha_1 f(x) + \alpha_2 g(x) \\ g(x+a) &= \beta_1 f(x) + \beta_2 g(x) \end{aligned}\right] \tag{6.7}$$

where α_1, α_2, β_1 and β_2 are the real functions of energy E. Another general solution to Eq. (6.3) as obtained from the functions (6.7) is

$$\begin{aligned} \psi(x + a) &= A\, f(x + a) + B\, g(x + a) \\ &= (A\alpha_1 + B\beta_1)\, f(x) + (A\alpha_2 + B\beta_2)\, g(x) \end{aligned} \tag{6.8}$$

If the constants A and B are chosen such that

$$\left.\begin{aligned} A\alpha_1 + B\beta_1 &= \lambda A \\ \text{and} \qquad A\alpha_2 + B\beta_2 &= \lambda B \end{aligned}\right] \tag{6.9}$$

where λ is a constant, we find from Eqs. (6.6) and (6.8) that the functions $\psi(x)$ has the property

$$\psi(x + a) = \lambda\psi(x) \tag{6.10}$$

The Eqs. (6.9) yield non-zero values for A and B only if the determinant of the coefficients vanishes, i.e. if

$$\begin{vmatrix} (\alpha_1 - \lambda) & \beta_1 \\ \alpha_2 & (\beta_2 - \lambda) \end{vmatrix} = 0 \tag{6.11}$$

or

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + \alpha_1\beta_2 - \alpha_2\beta_1 = 0 \tag{6.12}$$

We now show that $\alpha_1\beta_2 - \alpha_2\beta_1 = 1$ Since $g(x)$ and $f(x)$ are the real and independent solutions to Eq. (6.3), we get

$$\frac{d^2g(x)}{dx^2} + \frac{2m}{\hbar^2}\left[E - V(x)\right] g(x) = 0$$

and

$$\frac{d^2f(x)}{dx^2} + \frac{2m}{\hbar^2}\left[E - V(x)\right] f(x) = 0$$

Multiplying the former by $f(x)$ and the latter by $g(x)$ and subtracting, we obtain

$$f(x)\frac{d^2g(x)}{dx^2} - g(x)\frac{d^2f(x)}{dx^2} = 0$$

or $$f(x)\frac{dg(x)}{dx} - g(x)\frac{df(x)}{dx} = \text{constant}$$

The left hand side is called Wronskian, $W(x)$, of the solutions and is constant in this case. Further, from Eqs. (6.7), we obtain

$$f(x+a)\frac{dg(x+a)}{dx} - g(x+a)\frac{df(x+a)}{dx} = (\alpha_1\beta_2 - \alpha_2\beta_1)\left[f(x)\frac{dg(x)}{dx} - g(x)\frac{df(x)}{dx}\right]$$

or $$W(x+a) = (\alpha_1\beta_2 - \alpha_2\beta_1)\ W(x)$$

Hence

$$\alpha_1\beta_2 - \alpha_2\beta_1 = 1 \tag{6.13}$$

Thus, Eq. (6.12) becomes

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + 1 = 0 \tag{6.14}$$

It gives two possible values of λ and hence two functions $\psi_1(x)$ and $\psi_2(x)$ which have the property as given in Eq. (6.10). We consider the following two cases :

Case I *For energy ranges such that* $(\alpha_1 + \beta_2)^2 < 4$

The Eq. (6.14) will have complex roots λ_1 and λ_2. Since $\lambda_1\lambda_2 = 1$ these will be complex conjugates of each other. We may, therefore, write

$$\left.\begin{aligned} \lambda_1 &= e^{ika} \\ \text{and} \quad \lambda_2 &= e^{-ika} \end{aligned}\right] \tag{6.15}$$

where k is real. The corresponding functions $\psi_1(x)$ and $\psi_2(x)$ then have the property

$$\left.\begin{aligned} \psi_1\ (x + a) &= e^{ika}\ \psi_1(x) \\ \text{and} \quad \psi_2\ (x + a) &= e^{-ika}\ \psi_2(x) \end{aligned}\right] \tag{6.16}$$

or, in general,

$$\psi(x+a) = e^{\pm ika}\ \psi(x) \tag{6.17}$$

It is easy to see that a function having such a property is the Bloch function of the type of Eq. (6.4). To check this we replace x by $x+a$ in the Bloch function (6.4) and obtain

$$\psi(x+a) = e^{\pm ik(x+a)}\ u_k(x+a)$$

Using Eq. (6.5), it becomes

$$\psi(x+a) = e^{\pm ika}\ \psi(x)$$

which is the same as (6.17). Thus we find that the function $\psi(x)$, which represents a solution to Eq. (6.3), has the same property as that expressed by Eq. (6.17), Eq. (6.10) with $\lambda = e^{\pm ika}$, or the Bloch function (6.4). Hence

the Bloch functions of the type of Eq. (6.4) represent the solutions to the Schrodinger equation (6.13). This proves the Bloch theorem.

Case II *For the energy ranges such that* $(\alpha_1 + \beta_2)^2 > 4$

The Eq. (6.14) will have real roots λ_1 and λ_2 which may be taken as

$$\lambda_1 = e^{\mu a} \quad \text{and} \quad \lambda_2 = e^{-\mu a} \tag{6.18}$$

where μ is real. The corresponding solutions to the Schrodinger equation are

$$\psi_1(x) = e^{\mu x} u(x) \quad \text{and} \quad \psi_2(x) = e^{-\mu x}\, u(x) \tag{6.19}$$

Though mathematically valid, these solutions are not the acceptable electron wave functions because these are not bounded. The ratio $\dfrac{\psi(x+na)}{\psi(x)}$ approaches infinity as n approaches $\pm\infty$. This is contradictory to the requirement that the wave functions should remain finite at all points in space because of their probabilistic interpretation. In case I, however, $|\psi(x+na)|$ is equal to $|\psi(x)|$ and the ratio of these is always finite.

The roots (6.15) and (6.18) depend on α and β and hence are functions of energy E. The allowed roots (6.15) are associated with allowed energy regions and the disallowed roots (6.18) are associated with the forbidden energy regions. This means that the energy spectrum of an electron moving in a periodic potential consists of allowed and forbidden energy regions or bands. This result will be illustrated using the Kronig-Penny model.

In three dimensions, the Bloch theorem is expressed as

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}}\, u_k(\mathbf{r}) \tag{6.20}$$

If the lattice potential vanishes, $u_k(\mathbf{r})$ becomes constant and Eq. (6.20) gives

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}}$$

Thus the wave function becomes the one of a free electron.

6.3 THE KRONIG-PENNEY MODEL

This model illustrates the behaviour of electrons in a periodic potential by assuming a relatively simple one-dimensional model of periodic potential as shown in Fig. 6.2. The potential energy of an electron in a linear array of positive nuclei is assumed to have the form of a periodic array of square wells with period of $(a + b)$. At the bottom of a well, i.e. for $0 < x < a$, the electron is assumed to be in the vicinity of a nucleus and the potential energy is taken as zero, whereas outside a well, i.e. for $-b < x < 0$, the potential energy is assumed to be V_0. Although this model employs a very crude approximation to the type of periodic potential existing inside a lattice, yet it is very useful to illustrate various important features of the quantum behaviour of electrons in the periodic lattice. The wave functions

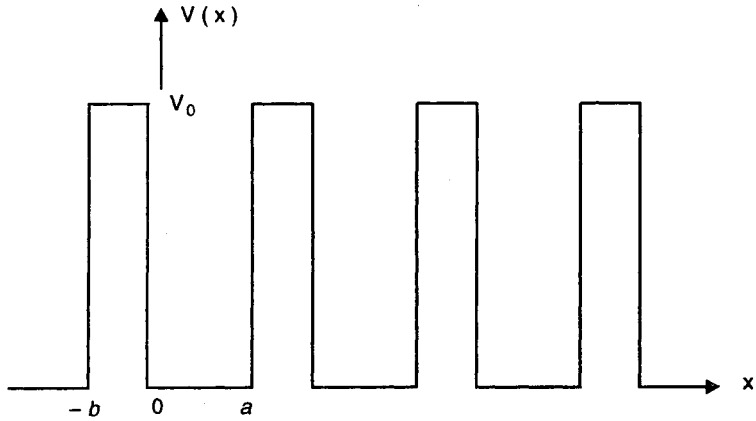


Fig. 6.2. *One-dimensional periodic potential used by Kronig and Penney.*

are obtained by writing the Schrodinger equations for the two regions as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0 \qquad \text{for } 0 < x < a \tag{6.21}$$

and

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V_0)\psi = 0 \qquad \text{for } -b < x < 0 \tag{6.22}$$

Assuming that the energy E of the electrons is less than V_0, we define two real quantities α and β as

$$\alpha^2 = \frac{2mE}{\hbar^2} \quad \text{and} \quad \beta^2 = \frac{2m(V_0 - E)}{\hbar^2} \tag{6.23}$$

Therefore, the Eqs. (6.21) and (6.22) become

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \qquad \text{for } 0 < x < a \tag{6.24}$$

and

$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \qquad \text{for } -b < x < 0 \tag{6.25}$$

Since the potential is periodic, the wave functions must be of the form of Bloch function (6.4), i.e.,

$$\psi(x) = e^{ikx}\, u_k(x) \tag{6.26}$$

where $u_k(x)$ is the periodic function in x with periodicity of $(a+b)$, i.e.,

$$u_k(x) = u_k(x + a + b) \tag{6.27}$$

From Eq. (6.26), we have

$$\frac{d\psi}{dx} = ike^{ikx}u_k(x) + e^{ikx}\frac{du_k}{dx}$$

and

$$\frac{d^2\psi}{dx^2} = -k^2e^{ikx}u_k(x) + 2ike^{ikx}\frac{du_k}{dx} + e^{ikx}\frac{d^2u_k}{dx^2} \tag{6.28}$$

Substituting Eqs. (6.26) and (6.28) in Eqs. (6.24) and (6.25), we get

$$\frac{d^2u_1}{dx^2} + 2ik\frac{du_1}{dx} + \left(\alpha^2 - k^2\right)u_1 = 0 \qquad 0 < x < a \tag{6.29}$$

and

$$\frac{d^2u_2}{dx^2} + 2ik\frac{du_2}{dx} - \left(\beta^2 + k^2\right)u_2 = 0; \qquad -b < x < 0 \tag{6.30}$$

where u_1 and u_2 represent the values of $u_k(x)$ in the intervals $0 < x < a$. and $-b < x < 0$ respectively. The general solutions to these equations are

$$u_1 = A\,e^{i(\alpha-k)x} + B\,e^{-i(\alpha+k)x}; \qquad 0 < x < a \tag{6.31}$$

and

$$u_2 = C\,e^{(\beta-ik)x} + D\,e^{-(\beta+ik)x}; \qquad -b < x < 0 \tag{6.32}$$

where A, B, C and D are constants which can be determined from the following boundary conditions :

$$u_1(0) = u_2(0); \qquad \left(\frac{du_1}{dx}\right)_{x=0} = \left(\frac{du_2}{dx}\right)_{x=0} \tag{6.33}$$

$$u_1(a) = u_2(-b); \qquad \left(\frac{du_1}{dx}\right)_{x=a} = \left(\frac{du_2}{dx}\right)_{x=-b} \tag{6.34}$$

The conditions (6.33) correspond to the requirements that the wave function ψ and its derivative $\frac{d\psi}{dx}$, and hence u and $\frac{du}{dx}$ must be continuous, whereas the conditions (6.34) correspond to the requirements of the periodicity of $u_k(x)$. Applying these boundary conditions to (6.31) and (6.32) we obtain the following four linear homogeneous equations :

$$\begin{aligned}
A + B &= C + D \\
Ai(\alpha-k) - Bi(\alpha+k) &= C(\beta-ik) - D(\beta+ik) \\
Ae^{i(\alpha-k)a} + Be^{-i(\alpha+k)a} &= Ce^{-(\beta-ik)b} + De^{(\beta+ik)b} \\
Ai(\alpha-k)e^{i(\alpha-k)a} - Bi(\alpha+k)e^{-i(\alpha+k)a} &= C(\beta-ik)e^{-(\beta-ik)b} - Di(\beta+ik)e^{(\beta+ik)b}
\end{aligned} \tag{6.35}$$

These equations are used to determine the constants A, B, C and D. A non-zero solution to these equations exists only if the determinant of the coefficients of A, B, C and D vanishes, i.e.,

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ i(\alpha-k) & -i(\alpha+k) & \beta-ik & -(\beta+ik) \\ e^{i(\alpha-k)a} & e^{-i(\alpha+k)a} & e^{-(\beta-ik)b} & e^{(\beta+ik)b} \\ i(\alpha-k)e^{i(\alpha-k)a} & -i(\alpha+k)e^{-i(\alpha+k)a} & (\beta-ik)e^{-(\beta-ik)b} & -(\beta+ik)e^{(\beta+ik)b} \end{vmatrix} = 0$$

On solving this determinant, we obtain

$$\frac{\beta^2+\alpha^2}{2\beta\alpha}\sinh\beta b \sin\alpha a + \cosh\beta b \cos\alpha a = \cos k(a+b) \tag{6.36}$$

To simplify this equation, Kronig and Penney considered the case when V_0 tends to infinity and b approaches zero but the product $V_0 b$ remains finite, i.e., the potential barriers become delta functions. Under these circumstances, the model is modified to the one of a series of wells separated by infinitely thin potential barriers. The quantity $\lim V_0 b$ for $V_0 \to \infty$ and $b \to 0$ is called the *barrier strength.*

As $b \to 0$, $\sinh \beta b \to \beta b$ and $\cosh \beta b \to 1$. Also, from Eqs. (6.23), we have

$$\frac{\beta^2+\alpha^2}{2\alpha\beta} = \frac{mV_0}{\alpha\beta\hbar^2}$$

Therefore, Eq. (6.36) becomes

$$\left(\frac{mV_0 b}{\hbar^2\alpha}\right)\sin\alpha a + \cos\alpha a = \cos ka \tag{6.37}$$

We define a quantity P as

$$P = \frac{mV_0 ba}{\hbar^2} \tag{6.38}$$

which is a measure of the area $V_0 b$ of the potential barrier. Thus increasing P has the physical meaning of binding an electron more strongly to a particular potential well. Using Eq. (6.38) in (6.37), we get

$$P\frac{\sin\alpha a}{\alpha a} + \cos\alpha a = \cos ka \tag{6.39}$$

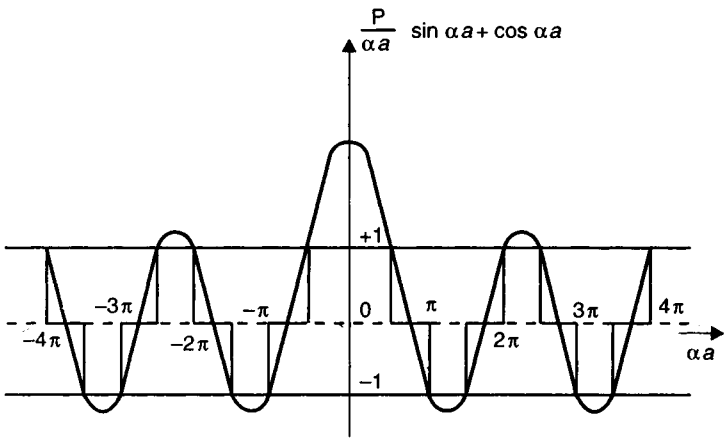


Fig. 6.3. *Plot of the left hand side of Eq. (6.39) as a function of αa using P = 3π/2. The solid and broken lines on the abscissa correspond to allowed and forbidden energy regions respectively.*

This is the condition which must be satisfied for the solutions to the wave equation to exit. Since cos*ka* lies between +1 and −1, the left hand side should assume only those values of αa for which its value lies between +1 and −1. Such values of αa, therefore, represent wave like solutions of the type of Eq. (6.26) and are allowed. The other values of αa are not allowed. Figure 6.3 shows a plot of the left hand side versus αa for $P = 3\pi/2$. The vertical axis lying between −1 and +1, as indicated by the horizontal lines, represents the values acceptable to the left hand side. It may be noted that since α^2 is proportional to the energy E, the abscissa is a measure of the energy. The following conclusions may be drawn from Fig. 6.3 :

(i) The energy spectrum of the electrons consists of alternate regions of allowed energy bands (solid lines on abscissa) and forbidden energy bands (broken lines).

(ii) The width of the allowed energy bands increases with αa or the energy.

(iii) The width of a particular allowed energy band decreases with increase in the value of P, i.e., with increase in the binding energy of the electrons. As $P \to \infty$, the allowed energy bands are compressed into energy levels and a line spectrum is resulted. In such a case, the Eq. (6.39) has solutions only if

$$\sin\alpha a = 0$$

or $$\alpha a = \pm n\pi$$

where n is an integer. Therefore, from Eq. (6.23), we have

$$E = \frac{\alpha^2\hbar^2}{2m} = \frac{\pi^2\hbar^2}{2ma^2}n^2 \tag{6.40}$$

This expression gives the energy levels of a particle in a constant potential box of atomic dimensions. This is what is expected physically as, for large P, the tunnelling through the barrier becomes difficult. In the other extreme case, i.e., when $P \to 0$, Eq. (6.39) becomes

$$\cos \alpha a = \cos ka$$

or $$\alpha = k$$

Therefore, Eq. (6.23) gives

$$E = \frac{k^2\hbar^2}{2m}$$

which is applicable to completely free electrons for which all the energies are allowed. This is again an expected result as, for P approaching zero, the electrons become free and, therefore, the free electron model involving quasi-continuous energy levels becomes applicable.

These results are summarized in Fig. 6.4 which shows the energy spectrum as a function of P. On the extreme left, i.e. for $P = 0$, the energy spectrum is quasi-continuous and on the extreme right, i.e. for $P = \infty$, a line spectrum is resulted. For any other value of P, the position and width of the allowed and forbidden bands are obtained by drawing a vertical line.

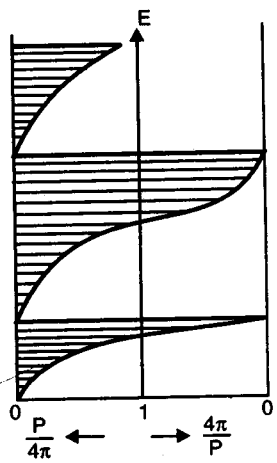


Fig. 6.4. *Allowed (shaded) and forbidden (open) energy ranges as a function of P.*

6.3.1 Energy versus Wave-Vector Relationship — Different Representations

It follows from Eq. (6.39) that coska takes a specific value for each allowed energy value E. However, coska is an even periodic function of k and remains unchanged if k is replaced by $k + 2\pi n/a$, where n is an integer. This means that the energy E is also an even periodic function of k with period of $2\pi/a$. This type of periodic representation of energy is shown in Fig. 6.5a for the first few allowed bands. This representation can be obtained by the periodic repetition of the region $-\pi/a < k < \pi/a$, i.e., the first *Brillouin*

zone, and is known as the *periodic or repeated zone scheme.* In this scheme, k is not uniquely defined as, for each value of E, there exist a number of k values. It is, therefore, generally convenient to introduce the following two schemes :

(*a*) The extended zone scheme

(*b*) The reduced zone scheme

A curve of E versus k in the extended zone scheme is shown by solid lines in Fig. 6.5b. The corresponding parabolic curve (dotted) for free electrons in the constant potential is also shown for comparison. We note that the E–k curve of the extended zone scheme is not continuous and has discontinuities at

$$k = \pm \frac{n\pi}{a} \tag{6.41}$$

where $n = 1, 2, 3, \ldots$ These values of k define the boundaries of the Brillouin zones. As described in Chapter II, the first Brillouin zone extends from $-\pi/a$ to $+\pi/a$, the second one extends from π/a to $2\pi/a$ and from $-\pi/a$ to $2\pi/a$, and so on. Since $k = 2\pi/\lambda$, Eq. (6.41) gives

$$n\lambda = 2a \tag{6.42}$$

which is the Bragg's law of reflection for normal incidence. It thus follows that, for the values of k as given by (6.41), the electrons suffer Bragg reflection which results in discontinuities in the E–k curve.

The E–k curve in the reduced zone scheme is shown in Fig. 6.5c It is obtained by 'reducing' the contents of the other zones so as to correspond, in general, to the first zone, i.e., to the region $-\pi/a \leq k \leq \pi/a$. The wave vector **k** belonging to this region is called the *reduced wave vector* and can be obtained from the wave vector **k** belonging to the region outside the first Brillouin zone by using the relation

$$\mathbf{k} = \mathbf{k}' - \mathbf{G} \tag{6.43}$$

where **G** is a suitably chosen reciprocal lattice vector.

6.3.2 Number of Wave Functions in a Band

It has been shown that, in an infinitely long one-dimensional crystal, there are certain allowed energy ranges which have a continuous distribution of energy. We now consider a finite crystal of length L and determine the number of possible wave functions, states or k-values per band. Using the periodic boundary conditions, we have

$$\psi(x + L) = \psi(x) \tag{6.44}$$

Since the wave functions are Bloch functions, it requires

$$e^{ik(x+L)}\, u_k(x+L) = e^{ikx}\, u_k(x)$$

Using Eq. (6.5), we obtain

$$e^{ik(x+L)} = e^{ikx}$$

or $$e^{ikL} = 1$$

which gives

$$k = \frac{2\pi n}{L} \tag{6.45}$$

with $n = \pm 1, \pm 2, \ldots$, or

$$k = \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \ldots \tag{6.46}$$

Therefore, the number of possible wave functions in the range dk is

$$dn = \frac{L}{2\pi} dk \tag{6.47}$$

Considering now the first Brillouin zone, the maximum value of k is π/a where a is the length of the primitive cell. If N is the number of primitive cells in the crystal of length L, then $a = L/N$. Therefore, the maximum value of k in the first Brillouin zone is $\frac{N\pi}{L}$. Thus the series (6.46) is cut off at N/L which means that the total number of allowed k-values in the first Brillouin zone is equal to N. Hence the total number of possible states in any energy band is equal to the number of primitive unit cells N. We further note that since each band can be occupied by a maximum of two electrons, each band can accommodate a maximum of $2N$ electrons. This result is quite important for explaining the distinction between metals, insulators and semiconductors.

6.4 VELOCITY AND EFFECTIVE MASS OF ELECTRON

6.4.1 Velocity of Electron

We consider the velocity of an electron described by a wave vector **k**. According to the wave mechanical theory of particles, a particle moving with a velocity v is equivalent to a wave packet moving with a group velocity v_g. Thus, we have

$$v = v_g = \frac{d\omega}{dk} \tag{6.48}$$

where ω is the angular frequency of the deBroglie waves. Also, the energy E of the particle is given by

$$E = \hbar\omega$$

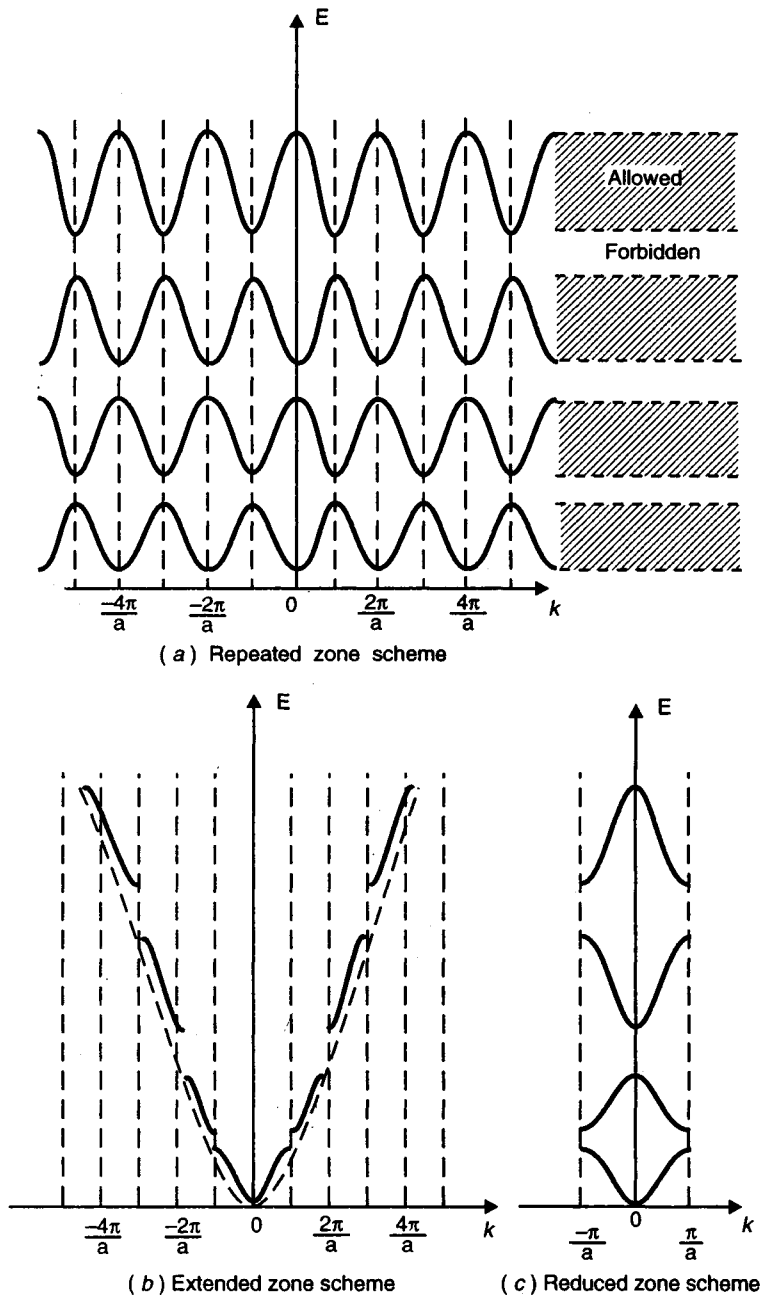


Fig. 6.5. *E versus k curves in three different zone schemes drawn on the same scale. Figure (b) also shows allowed and forbidden energy bands. The dotted parabolic curve in Fig. (b) corresponds to free electrons in a constant potential.*

Therefore, Eq. (6.48) can be written as

$$v = \frac{1}{\hbar}\frac{dE}{dk} \tag{6.49}$$

For free electrons, we have

$$E = \frac{\hbar^2 k^2}{2m}$$

and, therefore, Eq. (6.49) gives

$$v = \frac{\hbar k}{m} = \frac{p}{m}$$

This shows that, for a free electron, v is proportional to k. However, in the band theory, E is generally not proportional to k. The variation of E with k based on the band theory is shown in Fig. 6.6a. This shows that the slope dE/dk of the $E(k)$ curve is not constant but changes with k. Using this curve and employing Eq. (6.49), one can obtain v versus k curve as shown in Fig. 6.6b. This curve indicates that the velocity of the electron is zero for k=0 and $\pm\pi/a$ where the slope dE/dk is zero, i.e., at the top and bottom of the energy band (first Brillouin zone). For $k = k_0$, where k_o corresponds to the inflexion point of the $E(k)$ curve, the absolute value of the velocity attains a maximum value equal to the free electron velocity. Beyond the inflexion point, the velocity decreases with increase in energy which is altogether different from the behaviour of free electrons.

6.4.2 Effective Mass of Electron

Consider now the effect of applied electric field $\mathcal{E}$ on the motion of an electron present in the Brillouin zone. We assume only one electron in the Brillouin zone so that the Pauli's exclusion principle is not violated. If c is the velocity of the electron and the field $\mathcal{E}$ acts on it for a time dt, the increase in energy of the electron is given by

$$dE = e\mathcal{E}dx = e\mathcal{E}vdt = \frac{e\mathcal{E}}{\hbar}\left(\frac{dE}{dk}\right)dt \tag{6.50}$$

We can write

$$dE = \frac{dE}{dk}dk$$

$$\therefore \quad \frac{dE}{dk}dk = \frac{e\mathcal{E}}{\hbar}\left(\frac{dE}{dk}\right)dt$$

or

$$\frac{dk}{dt} = \frac{e\mathcal{E}}{\hbar} \tag{6.51}$$

or
$$\hbar \frac{dk}{dt} = \frac{dp}{dt} = F = e\mathcal{E} \tag{6.52}$$

where p represents the crystal momentum. It follows from Eq. (6.52) that the rate of change of crystal momentum is equal to the impressed force $e\mathcal{E}$. It is thus an analogue to Newton's second law for an electron in a periodic lattice.

The acceleration a of the electron is obtained by differentiating (6.49) with respect to t, i.e.,

$$a = \frac{dv}{dt} = \frac{1}{\hbar} \frac{d^2E}{dk^2} \left(\frac{dk}{dt} \right)$$

Using Eq. (6.51), we obtain

$$a = \frac{e\mathcal{E}}{\hbar^2} \frac{d^2E}{dk^2} \tag{6.53}$$

For a free electron of mass m,

$$a = \frac{e\mathcal{E}}{m} \tag{6.54}$$

A comparison of Eq. (6.53) with (6.54) yields an interesting result of the band theory, i.e., an electron moving in a crystal has an effective mass m* given by

$$m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2} \right)} \tag{6.55}$$

This indicates that the effective mass of an electron is not constant but depends on the value of d^2E/dk^2, i.e., on the shape of the $E(k)$ curve. The variation of the effective mass with k is shown in Fig. 6.6c. The effective mass is positive in the lower half of the energy band and negative in the upper half. It becomes infinite at the inflexion points of the $E(k)$ curve. Thus the electron behaves as a positively charged particle in the upper half of the band only.

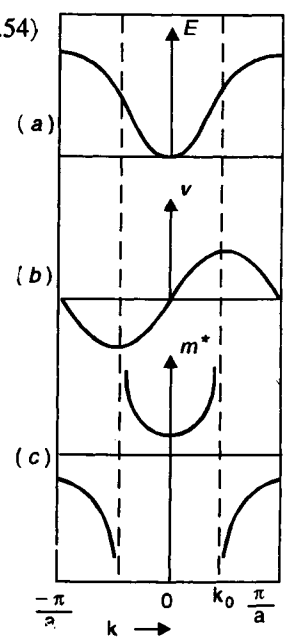


Fig. 6.6. *Variations of energy, velocity and effective mass with k. The dashed lines correspond to inflexion points in the E-k curve.*

It is generally convenient to introduce a factor

$$f_k = \frac{m}{m^*} = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)} \qquad (6.56)$$

which measures the extent up to which an electron in the k-state behaves as a free electron. If $f_k = 1$, the electron behaves totally as a free electron. If $f_k < 1$, i.e. $m^* > m$, the electron behaves as a heavy particle.

The concept of the negative effective mass may be understood in terms of the Bragg's reflection when k is close to $\pm\pi/a$. Due to the Bragg's reflection, a force applied in one direction leads to a gain of momentum in the opposite direction which results in the negative effective mass.

The concept of effective mass has a physical significance. It provides a satisfactory description of the charge carriers in crystals. In crystals such as alkali metals, which have partially filled energy band, the conduction takes place mainly through electrons. However, in crystals for which the energy band is nearly full, except for a few electron vacancies near the top of the band, these negative charge and negative mass vacancies may be considered as positive charge and positive mass particles called *holes* which act as positive charge carriers to produce conduction.

6.5 DISTINCTION BETWEEN METALS, INSULATORS AND SEMICONDUCTORS

To distinguish between metals, insulators and semiconductors on the basis of the band theory, we consider an energy band which is filled with electrons up to a certain value k_1 ($k_1 < \pi/a$) as shown in Fig. 6.7. It is of interest to know the effective number of free electrons present in this band which gives information about the conductivity of the band. In the expression (6.56), we introduced the term f_k that represents the extent up to which an electron in the k-state behaves as a free electron and hence participates in electrical conduction. Thus, in order to determine the total effective number of free electrons in the energy band, we have to sum up over all the possible f_k's corresponding to the various occupied energy states present in the band. Therefore, we can write

$$N_{eff} = \Sigma f_k \qquad (6.57)$$

The Eq. (6.47) gives the number of possible states in the interval dk for a one-dimensional lattice of length L as

$$dn = \frac{L}{2\pi} dk \qquad (6.47)$$

Since each state is occupied by a maximum of two electrons, the effective number of free electrons present in the band within the limits $-k_1$ and $+k_1$ be-

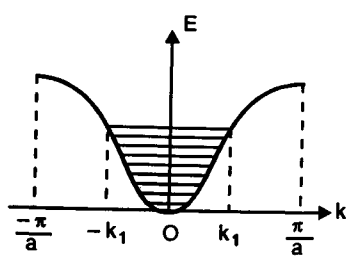


Fig. 6.7. *Energy band (first Brillouin zone) filled up to the state k_1.*

comes

$$N_{eff} = 2\frac{L}{2\pi}\int_{-k_1}^{k_1} f_k dk = \frac{L}{\pi}\int_{-k_1}^{k_1} f_k dk \tag{6.58}$$

Using Eq. (6.56), we obtain

$$N_{eff} = \frac{2L}{\pi}\frac{m}{\hbar^2}\int_0^{k_1}\frac{d^2E}{dk^2}dk$$

$$= \frac{2Lm}{\pi\hbar^2}\left(\frac{dE}{dk}\right)_{k=k_1} \tag{6.59}$$

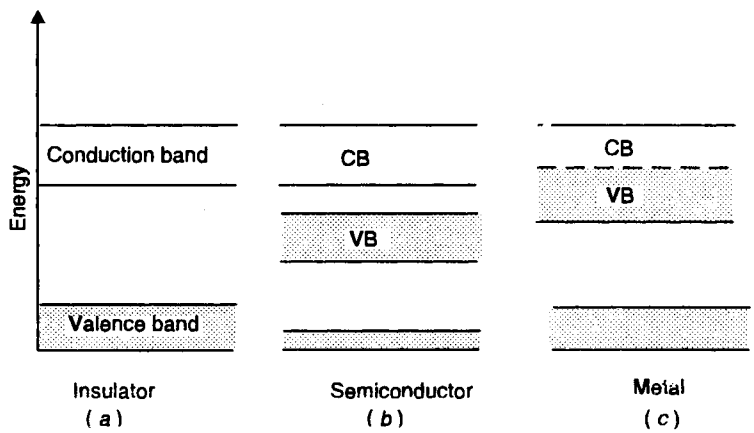


Fig. 6.8. *Electron distribution in an insulator, intrinsic semiconductor and metal at* 0 K. *The shaded regions are occupied by electrons.*

This yields the following results :

(i) The effective number of free electrons in a completely filled band is zero. This is because *dE/dk* vanishes at the top of the band.

(ii) The effective number of electrons attains a maximum value when *dE/dk* becomes maximum, i.e., when the band is filled up to the point of inflexion.

It follows from the above discussion that a solid having a certain number of energy bands completely filled and all other bands completely empty, as shown in Fig. 6.8a, is an insulator. On the other hand, a solid having a partly filled energy band has a metallic character (Fig. 6.8c). The situation depicted in Fig. 6.8a can strictly occur only at 0 K. At temperatures greater than 0 K, some electrons from the topmost filled energy band (valence band)

may get excited to the next higher empty band (conduction band) where they participate in the conduction process. The number of such free or conduction electrons depends on the value of the forbidden energy gap between the valence band and the conduction band. The larger the band gap, the smaller the number of free electrons and hence the larger the tendency of the material to behave as an insulator for all practical purposes. An example of this type of material is diamond which has the band gap of about 6 eV. If the band gap is small ($\cong$1eV) the number of thermally excited electrons becomes appreciable even at ordinary temperatures and the material behaves as an intrinsic semiconductor (Fig. 6.8b). The examples are germanium and silicon. At 0 K, even these materials behave as insulators because electrons from the valence band cannot be thermally excited to the conduction band to cause conduction. Thus it is evident that all the intrinsic semiconductors are insulators at 0 K and all the insulators may behave as semiconductors at temperatures sufficiently higher than 0 K.

It is important to note that the conductivity of semiconductors increases with temperature owing to the increase in the number of conduction band electrons. The conductivity of metals, however, decreases with increase in temperature. This is because more and more phonons are excited at higher temperatures which may scatter conduction electrons and reduce their mobility.

SOLVED EXAMPLES

Example 6.1. Using the Kronig-Penney model, show that for $P << 1$, the energy of the lowest energy band is

$$E = \frac{\hbar^2 P}{ma^2}$$

Solution. Referring to Eq. (6.39), the energy of the lowest band corresponds to $k = \pm \pi/a$, i.e., when

$$P\left(\frac{\sin\alpha a}{\alpha a}\right) + \cos\alpha a = \pm 1$$

Considering only the magnitude on the right hand side, we obtain

$$\frac{P}{\alpha a}(\sin\alpha a) = 1 - \cos\alpha a$$

or

$$\frac{2P}{\alpha a}\sin\left(\frac{\alpha a}{2}\right)\cos\left(\frac{\alpha a}{2}\right) = 2\sin^2\left(\frac{\alpha a}{2}\right)$$

For $P << 1$, we can write

$$\tan\left(\frac{\alpha a}{2}\right) = \frac{P}{\alpha a} = \tan\left(\frac{P}{\alpha a}\right)$$

or $$\frac{\alpha a}{2} = \frac{P}{\alpha a}$$

or $$\alpha^2 = \frac{2P}{a^2}$$

Using Eq. (6.23), we get

$$\frac{2mE}{\hbar^2} = \frac{2P}{a^2}$$

or $$E = \frac{\hbar^2 P}{ma^2}$$

Example 6.2. The energy near the valence band edge of a crystal is given by

$$E = -Ak^2$$

where $A = 10^{-39}$ Jm^2. An electron with wave vector $\mathbf{k} = 10^{10}\,\mathbf{k}_x$ m^{-1} is removed from an orbital in the completely filled valence band. Determine the effective mass, velocity, momentum and energy of the hole.

Solution. We have

$$E = -Ak^2$$

$\therefore$ $$\frac{dE}{dk} = -2Ak,$$

and $$\frac{d^2E}{dk^2} = -2A$$

The effective mass of the electron is

$$m_e^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)}$$

$$= -\frac{\hbar^2}{2A} = -\frac{\left(1.05\times10^{-34}\right)^2}{2\times10^{-39}} = -5.5\times10^{-30}\ \text{kg}.$$

Since the effective mass of a hole is opposite to that of an electron at the same location in the energy band, the effective mass of hole is

$$m_h^* = -m_e^* = 5.5\times10^{-30}\,\text{kg}$$

The wave vector of hole is opposite to that of electron, i.e.,

$$\mathbf{k}_h = -\mathbf{k}_e = -10^{10}\,\hat{\mathbf{k}}_x\ \text{m}^{-1}$$

The momentum of hole is

$$\mathbf{p}_h = \hbar\mathbf{k}_h$$

$$= -1.05\times10^{-34}\times10^{10}\,\hat{\mathbf{k}}_x = -1.05\times10^{-24}\,\hat{\mathbf{k}}_x\ \text{Jsm}^{-1}$$

The velocity of hole is

$$v_h = \frac{\mathbf{p}_h}{m_h^*} = -\frac{1.05\times10^{-24}\,\hat{\mathbf{k}}_x}{5.5\times10^{-30}} = -1.9\times10^5\,\hat{\mathbf{k}}_x\ \text{ms}^{-1}$$

The energy of the electron with wave vector $\mathbf{k}_e$ is

$$E_e = -Ak^2{}_e$$

$$= -10^{-39}(10^{10}\,\hat{\mathbf{k}}_x)^2$$

$$= -10^{-19}\ \text{J}$$

Therefore, the energy of the hole referred to zero at the top of the valence band is

$$E_h = -E_e$$

$$= 10^{-19}\ \text{J}$$

SUMMARY

1. The band theory assumes that the potential experienced by an electron in a crystal is periodic with period equal to the lattice constant.

2. According to the Bloch theorem, the solution to the Schrodinger wave equation for an electron moving in a periodic potential is of the form of the Bloch function, i.e.,

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}}\,u_k(\mathbf{r})$$

where $u_k(\mathbf{r})$ is invariant under a crystal lattice translation.

3. The energy spectrum of an electron moving in a periodic potential consists of a set of allowed energy regions or bands which are separated by forbidden energy regions called the band gaps. The forbidden energy regions are those in which **k** assumes complex values and no Bloch function type solutions to the wave equation are possible.

4. The width of the allowed energy bands increases with increase in total energy and decrease in binding energy of the electrons. For very large binding energies, the energy bands are transformed into energy levels. If the electrons are completely free, the energy spectrum is quasi-continuous.

5. The *E-k* curve in the extended zone scheme has discontinuities at

$$k = \pm\frac{n\pi}{a}$$

where a is the length of the primitive cell and $n = 1, 2, 3, \ldots$ These k-values define the boundaries of the Brillouin zones which produce Bragg reflection of electrons.

6. In a crystal having N primitive cells, the maximum number of energy states per band is N and the maximum number of electrons per band is $2N$.

7. The velocity of an electron is given by the expression

$$v = \frac{1}{\hbar}\frac{dE}{dk}$$

It becomes maximum at the inflextion points of the *E–k* curve where it approaches the free electron velocity. Beyond the inflexion points, the velocity decreases with increase in energy.

8. The effective mass of an electron moving in a crystal depends on the shape of the *E–k* curve and is given by the expression

$$m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)}$$

In the first Brillouin zone, the effective mass is positive in the lower half and negative in the upper half of the energy band. It beeomes infinite at the inflexion points of the *E–k* curve.

9. For a one-dimensional lattice of length L, the total effective number of free electrons in an energy band filled with electrons up to a certain value k_1 ($k_1 < \pi/a$) is

$$N_{eff} = \frac{2Lm}{\pi\hbar^2}\left(\frac{dE}{dk}\right)_{k=k_1}$$

where m is the mass of an electron. N_{eff} is zero at the top of the band and attains a maximum value at the points of inflexion.

VERY SHORT QUESTIONS

1. What is the Bloch function?
2. What is forbidden energy gap?
3. Can an electron possess negative effective mass? Justify.
4. Give the order of band gap for a metal, a semiconductor and an insulator.
5. Why a solid whose energy bands is filled can not be a metal?

SHORT QUESTIONS

1. What are the shortcomings of the free electron theory? What is the main cause of its failure?
2. Compare the salient features of the band theory vis-a-vis the free electron theory. Discuss the successes and failures of these theories.
3. What is the Bloch theorem ?
4. Explain the concept of forbidden energy bands.
5. Distinguish between a metal, a semiconductor and an insulator on the basis of their energy band structure.
6. Show that the existence of energy gaps at the boundaries of the Brillouin zone of a one-dimensional lattice is equivalent to the condition of Bragg reflection of electron waves.
7. Show the dependence of velocity of electron on wave vector **k** as predicted by the band theory.
8. Prove that the number of different **k**-states in the first Brillouin zone of a simple cubic lattice is equal to the number of lattice sites.
9. Prove that the total number of possible states in an energy band of a finite crystal is equal to the number of primitive cells in it.
10. How does the band theory lead to the concept of negative effective mass ?
11. The conductivity of metals decreases while that of semiconductors increases with rise in temperature. Explain.

LONG QUESTIONS

1. State and prove the Bloch theorem. Discuss its importance in the band theory.
2. Discuss the formation of allowed and forbidden energy bands on the basis of the Kronig-Penney model. Discuss the extreme conditions when energy levels are either discrete or continuous. What is the effect of changing the binding energy of electron on the energy bands ?
3. Prove that the motion of electrons through the periodic potential in solids gives rise to the band structure.
4. What are Bloch functions? Explain the origin of allowed and forbidden bands for electrons in solids. What is the number of orbitals in an energy band ?
5. Describe the periodic zone scheme, extended zone scheme and reduced zone scheme for representing E-k relationships.
6. Discuss the motion of electrons in one-dimension according to the band theory and show the variation of energy, velocity and effective mass as a function of wave-vector.
7. What is meant by the effective mass of an electron? What is its significance? Show that the effective mass of an electron in a crystal is inversely proportional to the second derivative of the E-k curve. Discuss the conditions when the effective mass of an electron becomes positive, negative and infinity.

PROBLEMS

1. The potential of an electron in a one-dimensional lattice is of the same type as that used in the Kronig-Penney model. Assuming

$$V_0\, ab << \hbar^2/m,$$

prove that the energy band gap at $k = \pi/a$ is $2V_0 b/a$.

2. A one-dimensional lattice of spacing a has a potential distribution of the type as considered in the Kronig-Penney model. The value of the potential is $-V$ at each lattice point and abruptly changes to zero at a distance of $0.1a$ on either side of the lattice point. Determine the width of the first energy gap in the electron energy spectrum.

(0.37 V)

CHAPTER - VII

SEMICONDUCTORS

7.1 INTRODUCTION

Semiconductors are materials which have electrical conductivities lying between those of good conductors and insulators. The resistivity of semiconductors varies from 10^{-5} to 10^{+4} ohm-m as compared to the values ranging from 10^{-8} to 10^{-6} ohm-m for conductors and from 10^{7} to 10^{8} ohm-m for insulators. There are elemental semiconductors such as germanium and silicon which belong to Group IV of the periodic table and have resistivity of about 0.6 and 1.5×10^{3} ohm-m respectively. Besides these, there are certain compound semiconductors such as gallium arsenide (GaAs), indium phosphide (InP), cadmium sulphide (CdS), etc. which are formed from the combinations of the elements of Groups III and V, or Groups II and VI. Another important characteristic of the semiconductors is that they have small band gap. The band gap of semiconductors varies from 0.2 to 2.5 eV which is quite small as compared to that of insulators. The band gap of a typical insulator such as diamond is about 6 eV. This property determines the wavelength of radiation which can be emitted or absorbed by the semiconductor and hence helps to construct devices such as light emitting diodes (LEDs) and lasers. All the semiconductors have negative temperature coefficient of resistance. The band gap energies for the elements of Group IV at 0 K are as follows :

C (diamond)	5.51 eV
Ge	0.75 eV
Si	1.16 eV
Sn (grey)	0.08 eV
Pd	$\cong 0$

At room temperature, diamond behaves as an insulator, Ge and Si as semiconductors and Sn and Pd as conductors.

The importance of semiconductors is further increased due to the fact that the conductivity and the effective band gaps of these materials can be modified by the introduction of impurities which strongly affect their elec-

tronic and optical properties. The process of introduction of impurities in semiconductors in a precisely controlled manner is called *doping*. Depending on the nature of impurities added, the semiconductors are classified as follows :

(a) Pure or intrinsic semiconductors

(b) Impurity or extrinsic semiconductors

The intrinsic semiconductors are pure semiconductors in which no impurity atoms are added, whereas the extrinsic semiconductors are doped semiconductors in which suitable impurity atoms are added to modify the properties. In the present chapter, the effects of impurities and charge carrier concentrations in semiconductors are discussed.

7.2 PURE OR INTRINSIC SEMICONDUCTORS

As stated above, the intrinsic semiconductors such as pure Ge or Si are undoped semiconductors. The electrical conductivity of this type of semicondutor is solely determined by thermally generated carriers. To understand the mechanism of conduction, we consider the bonding between

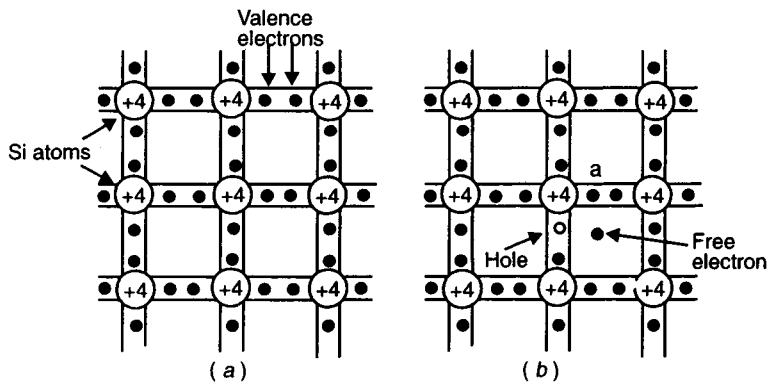


Fig. 7.1. *(a) A two-dimensional representation of Si crystal (b) Silicon crystal containing an electron-hole pair.*

atoms in these semiconductors. Consider, for example, the case of silicon with atomic number 14. Each silicon atom has four valence electrons and can form four covalent bonds with four neighbouring silicon atoms which are directed along the corners of a regular tetrahedron. The silicon crystal, therefore, exhibits a three-dimensional regular network type structure which, for simplicity, is represented by a two-dimensional network as shown in Fig. 7.1a. Apparently, all the valence electrons in a silicon crystal participate in the formation of covalent bonds and no electron is free to cause conduction

particularly at 0 K. Hence pure silicon behaves as an insulator at 0 K. As the temperature increases above 0 K, some of the valence electrons may acquire sufficient thermal energy to break their covalent bonds and become free from the influence of cores of the atoms. These electrons move randomly in the crystal and are referred to as the *conduction electrons.* Each escaped electron leaves behind an empty space called a *hole* which also acts as a current carrier. Thus when a valence electron breaks away from a covalent bond, an electorn-hole pair is generated and two carriers of electricity are produced as shown in Fig. 7.1b. When a valence electron located adjacent to a hole acquires sufficient thermal energy, it may jump into the hole position to reconstruct the broken covalent bond and a hole is created at the initial position of the electron. Thus the motion of an electron may also be regarded as the motion of a hole in the opposite direction. These electrons and holes move in opposite directions under the effect of an external electric field and constitute the current.

The energy band diagram of the intrinsic semiconductor is shown in Fig. 7.2. At 0 K, the valence band is completely filled and the conduction band is completely empty. The semiconductor, therefore, behaves as an insulator. The electrons present in the valence band do not conduct as these are bound to their respective cores. As temperature increases, some of the valence band electrons acquire sufficient thermal energy to jump to the conduction band leaving behind an equal number of holes in the valence band. The electrons in the conduction band and holes in the valence band behave as free carriers and increase the conductivity of the material. The conditions for the movement of electrons are, however, different from the conditions for movement of holes; the electrons move when the conduction band is nearly empty and the holes move when the valence band is nearly full. Thus the electrons move mainly under the influence of the applied field while the holes move under the combined effect of the applied electric field and the ionic field of the lattice. Thus the properties such as effective mass, mobility[1], etc. of a hole are quite different from the corresponding properties of electrons. For example, a hole has larger effective mass and lower mobility than the corresponding values for an electron. The charge of a hole is equal and opposite to that of an electron. These properties have been described in the previous chapter. It is now apparent that, in an intrinsic semiconductor, the number of electrons, n_i, in the conduction band is always equal to the number of holes, p_i, in the valence band, i.e., $n_i = p_i$, and either one of these is called the *intrinsic carrier concentration.*

Besides the generation of free electron-hole pairs, there is another process called *recombination* of carriers in semiconductors. A free electron

1. Mobility of a carrier is the velocity acquired by it in a unit electric field.

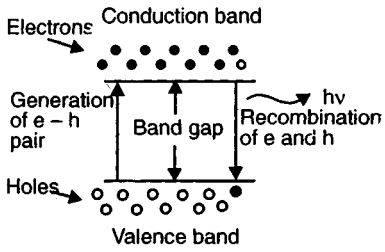


Fig. 7.2. *Energy band diagram of an intrinsic semiconductor showing generation and recombination of an electron and a hole.*

moving randomly in a semiconductor may encounter a hole and combine with it so as to reconstruct the broken covalent bond. Thus the electron-hole pair is destroyed and the free electron is converted into the bound electron. This recombination process is equivalent to an electron jumping from the conduction band to the valence band and occurs with the release of energy equal to the band gap energy in the form of electromagnetic radiation as shown in Fig. 7.2. In an intrinsic semiconductor, the rate of generation of carriers, g, depends on the temperature and nature of the material. The recombination rate, R, on the other hand, depends on the concentration of electrons and holes in a material at that temperature. In equilibrium, the generation rate just equals the recombination rate, i.e.,

$$g = R = C\, n_i p_i$$

where C is a proportionality constant which depends on the nature of the material. Since in an intrinsic semiconductor, $n_i = p_i$, we have

$$g = R = C\, n^2{}_i \qquad (7.1)$$

For a given temperature, the quantity n_i is a constant and depends only on the nature of the semiconductor. It will be realized later that the constant C is related to the densities of states of electrons and holes at the conduction band and valence band edges respectively.

7.3 IMPURITY OR EXTRINSIC SEMICONDUCTORS

The intrinsic semiconductors, as such, are of little importance owing to their very small and fixed conductivity. The introduction of impurity atoms, i.e. doping, is the most efficient and convenient method of increasing and altering the conductivity of an intrinsic semiconductor. Depending on the type of doping, excess electrons or holes are generated in the material which are free to conduct electricity. The impurity atoms frequently employed to dope pure silicon or germanium are the elements of Group III and Group V of the periodic table. These impurity atoms are referred to as acceptor or p-type and donor or n-type impurities as they contribute excess holes and electrons respectively to the semiconducting material. The semiconductor is accordingly known as p-type or n-type semiconductor. The dopants are added in the ratio of about 1 in 10^6 to 10^8 atoms of the semiconducting material. Such a small quantity of dopants does not bring about any structural changes in the semiconductor as the impurity atoms

replace the regular atoms in the crystal. However, the conductivity of the semiconductors is greatly affected by such substitutions.

7.3.1 Donor or n-type Semiconductor

When a pentavalent impurity atom of Group V, such as phosphorus, arsenic or antimony, is introduced into silicon, four of its five valence electrons form covalent bonds with the neighbouring four silicon atoms while the fifth valence electron remains loosely bound to its nucleus as shown in Fig. 7.3a. A small but definite amount of energy is required to detach this fifth electron from its nucleus and make it free to conduct. The energy required is, however, quite small as compared to the energy required for breaking a covalent bond and can be easily provided by thermal agitation inside the crystal.

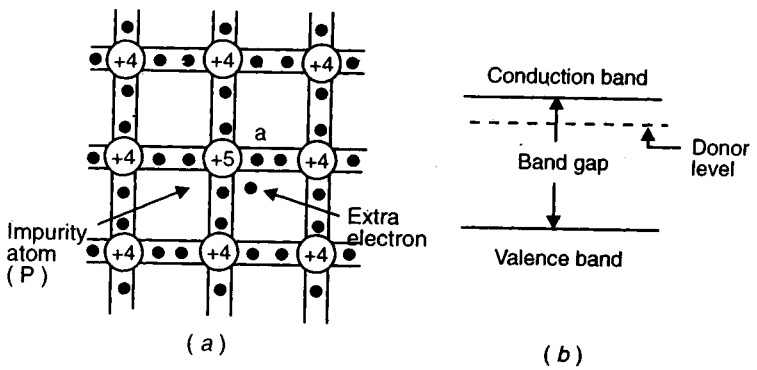


Fig. 7.3.*(a) A pentavalent impurity atom (P) in a silicon crystal. (b) Energy level diagram of an n-type semiconductor.*

The energy level corresponding to the fifth valence electron lies in the band gap just below the conduction band edge as shown in Fig. 7.3b. This level is called the *donor level.* The depth of the donor level below the conduction band is merely about 0.01 eV for Ge and 0.03 eV for Si. The electrons are, therefore, easily transferred to the conduction band leaving behind positively charged immobile impurity ions. Thus each pentavalent impurity atom donates one free electron to the semiconductor. Such impurities are, therefore, known as *donors or n-type impurities* and the semiconductor containing such impurity atoms is known as an *n-type semiconductor.* In these semiconductors, the current is carried mainly by electrons which are called *majority carriers.* The thermally generated holes are called *minority carriers.* The electron concentration, n, is obviously quite large as compared to hole concentration, p, but their product always remains constant.

i.e., $$np = n_i p_i = n^2_i \qquad (7.2)$$

where n_i and p_i are the intrinsic values of the carrier concentration. This relationship is called the *law of mass-action* and will be derived later.

7.3.2 Acceptor or p-type Semiconductor

If a trivalent impurity atom of Group III, such as boron, aluminium, gallium or indium, is introduced into silicon, it forms three covalent bonds with the neighbouring three silicon atoms while the fourth bond is not completed due to the deficiency of one electron. This incomplete bond is shown by broken line in Fig. 7.4a where the small circle (marked 'a') represents the electron deficiency. Thus the trivalent impurity atom has a tendency to accept one electron (say 'b') from a neighbouring silicon atom to complete the fourth covalent bond. This process requires a small amount of energy which is easily provided by the thermal agitation in the crystal. The transferred electron leaves behind a broken covalent bond, i.e., a hole, at position 'b' on the silicon atom which acts as a current carrier.

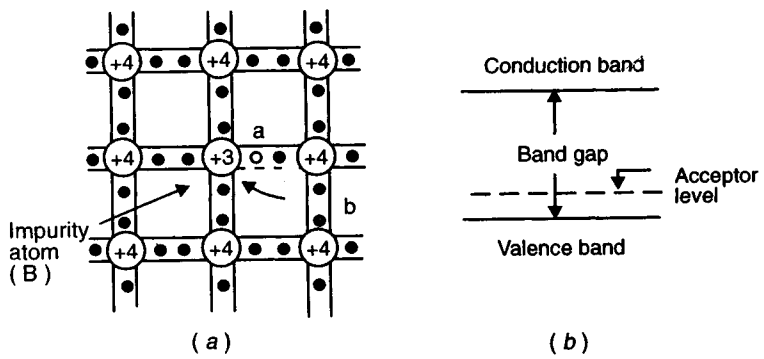


Fig. 7.4. *(a) A trivalent impurity atom (B) in a silicon crystal. (b) Energy level diagram of a p-type semiconductor.*

The energy level corresponding to the electron deficiency of the type 'a' is located just above the valence band and is called the *acceptor level.* The acceptor levels are located at a distance of about 0.01 eV above the top of the valence band in Ge and at about 0.046 to 0.16 eV in Si. An electron can be easily transferred from the valence band to the acceptor level by providing this small amount of energy. This creates a hole in the valence band which acts as a mobile current carrier. The negatively charged impurity atom, however, remains immobile and does not contribute to conduction. Thus each trivalent impurity atom can accept an electron from a neighbouring silicon atom to produce a hole in the semiconductor. Such impurities are, therefore, known as *acceptors or p-type impurities* and the semiconductor containing such impurity atoms is known as a *p-type semiconductor.* In these semicon-